# Topic: Sinusoidal Alternating Waveforms

## Book reference:

1. Alternating Current Circuits by G.F.Corcoran
2. Fundamental-of-Electric Circuits by Alexander
3. Introductory-circuit-analysis by Boyelstate

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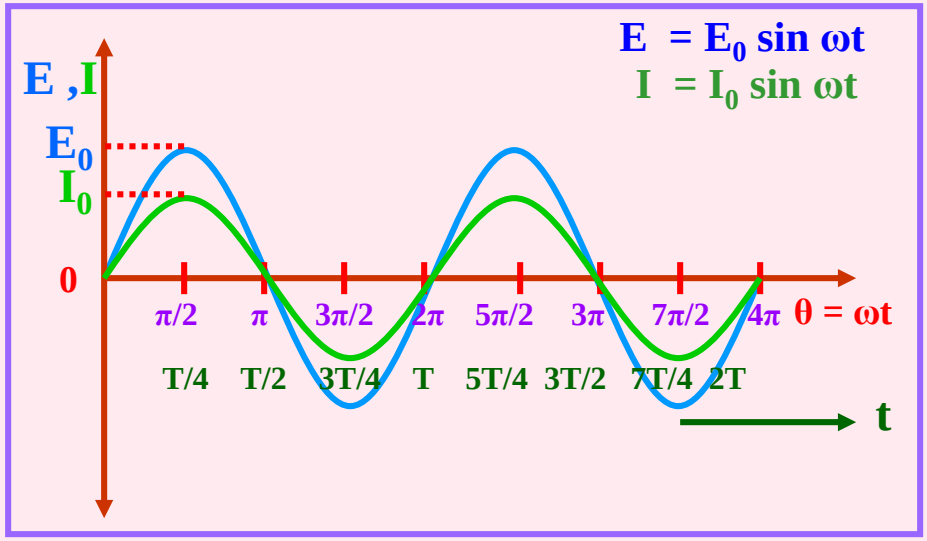
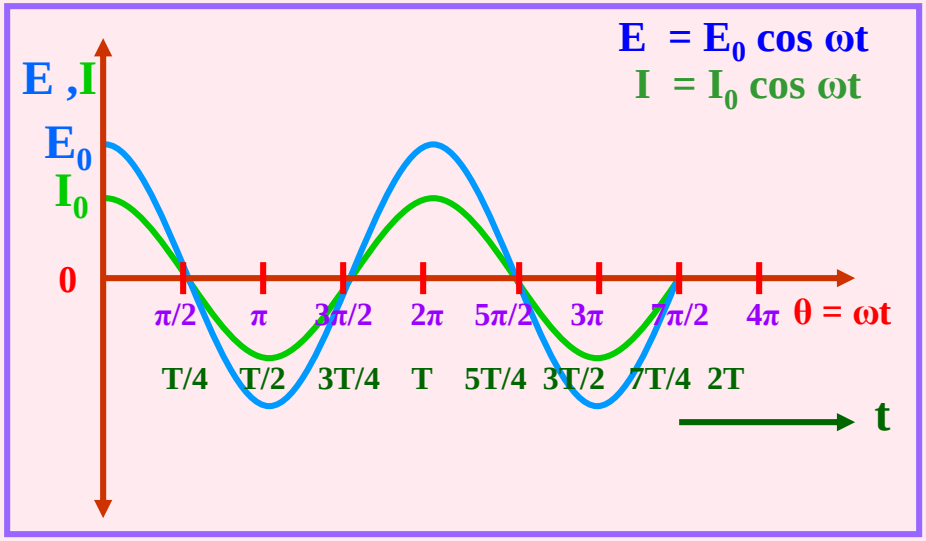
# Alternating emf:

**Alternating emf is that emf which continuously changes in magnitude and periodically reverses its direction.**

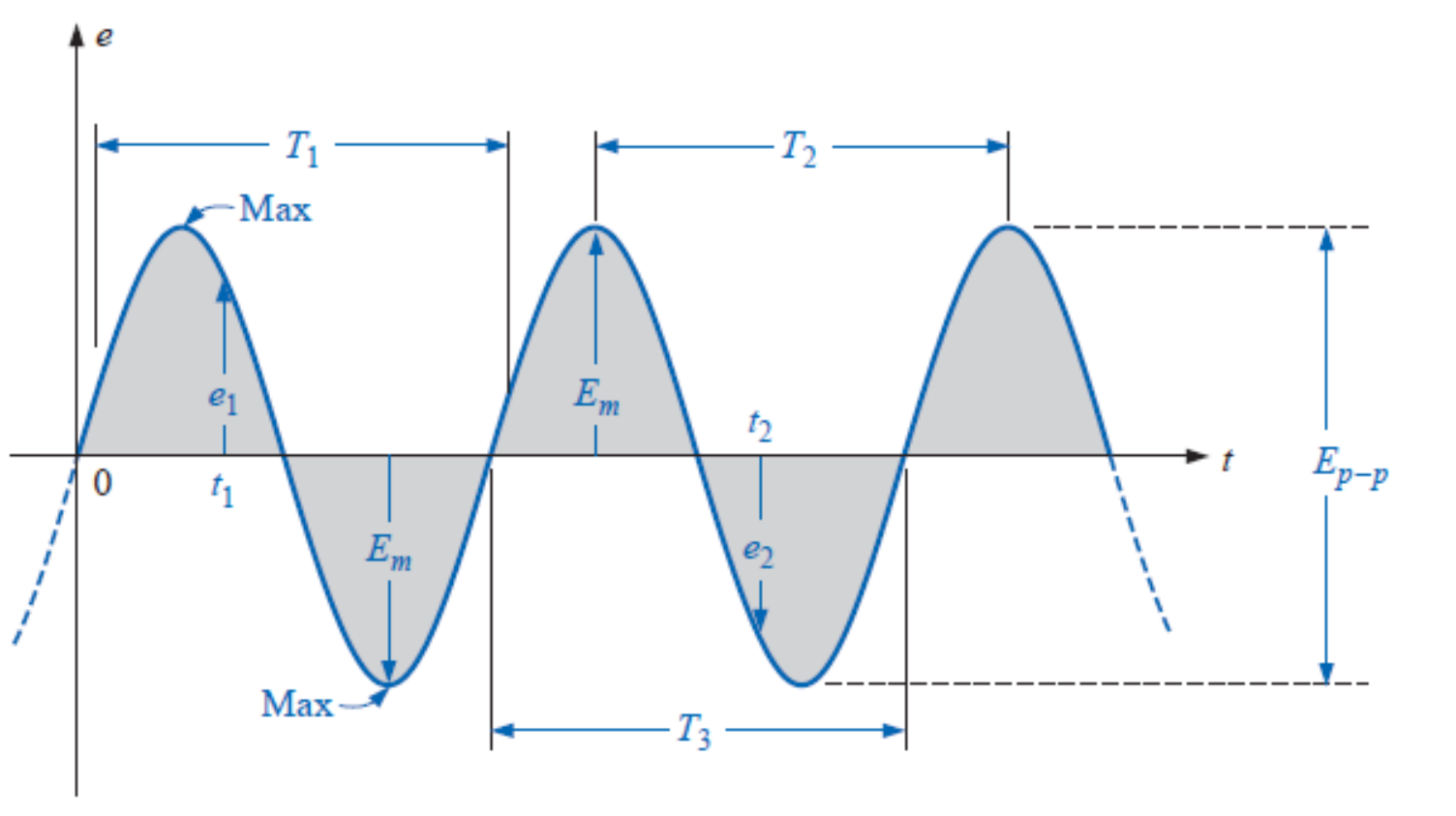
## Alternating Current:

**Alternating current is that current which continuously changes in magnitude and periodically reverses its direction.**

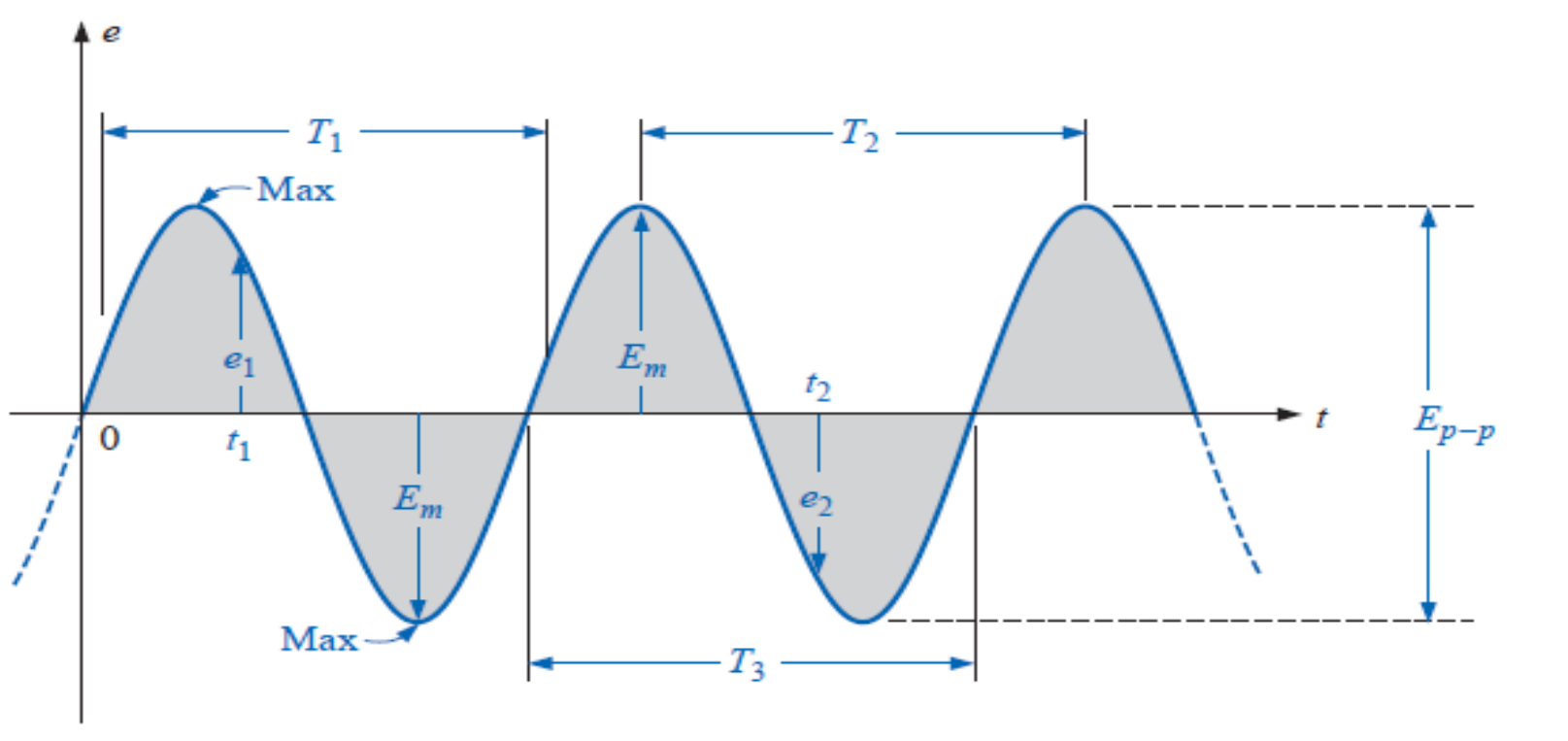


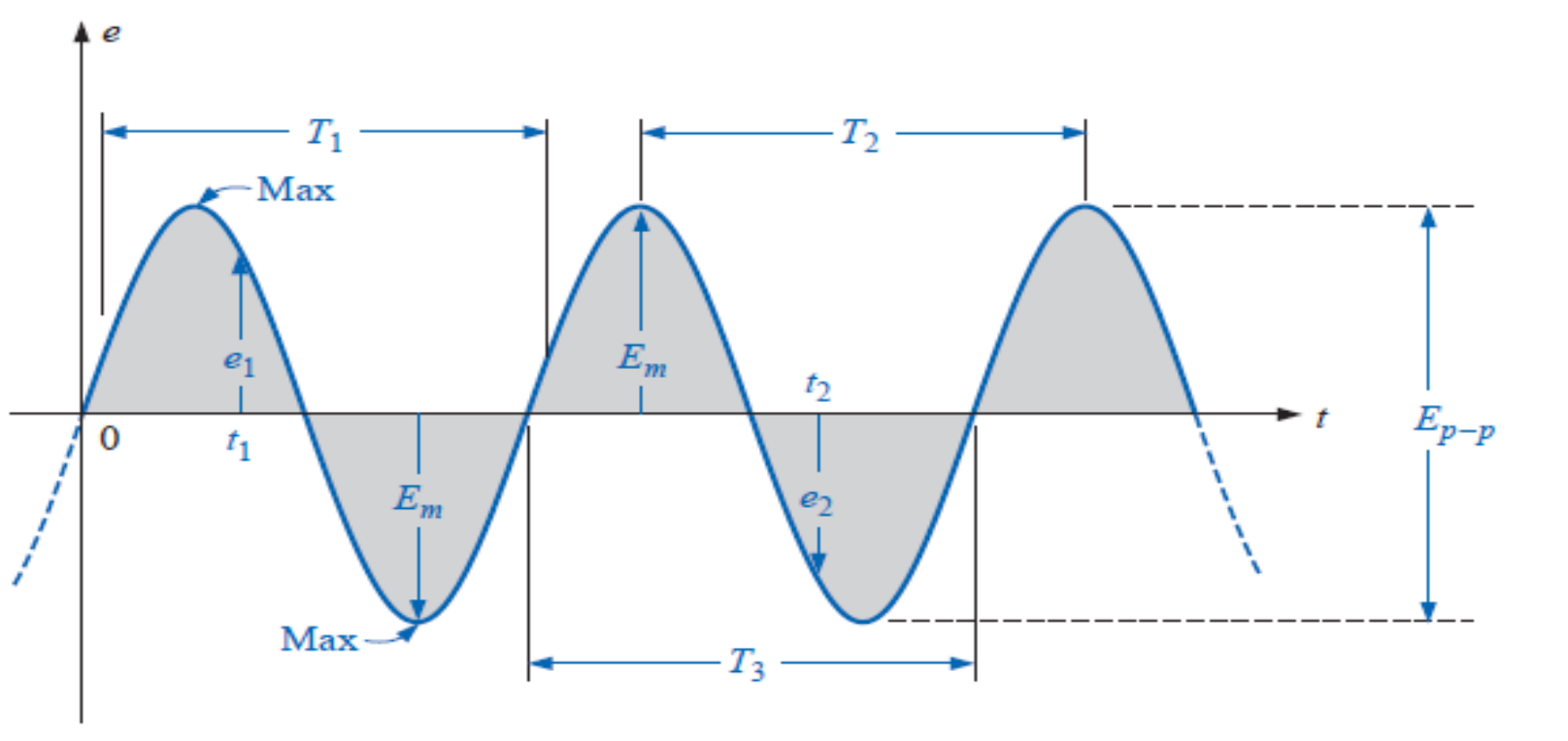
Instantaneous value: The magnitude of a waveform at any instant of time is known as instantaneous value ; denoted by lowercase letters (e1, e2).



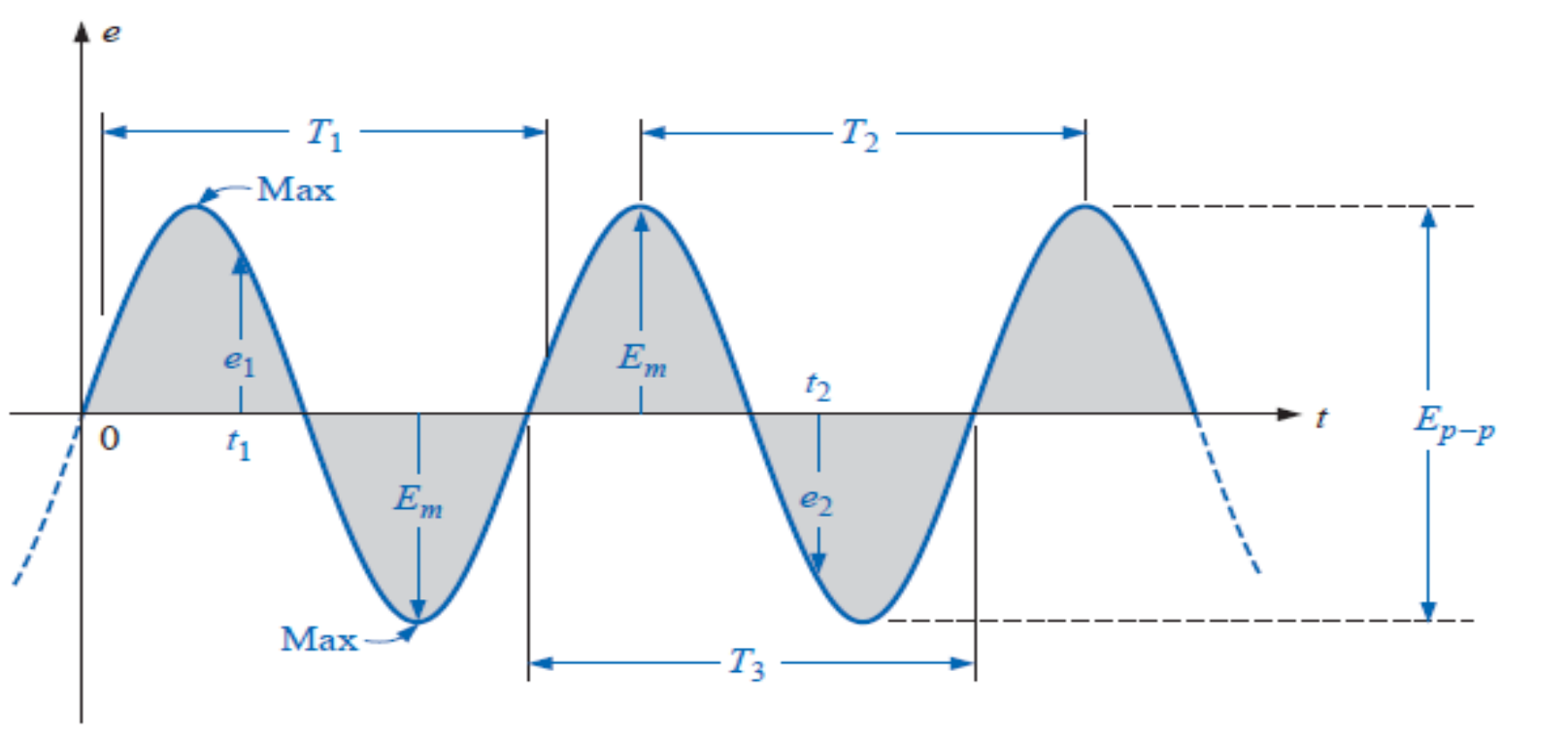
Peak amplitude: The maximum value of a waveform is known as peak amplitude value ; denoted by uppercase letters (Em for sources of voltage and Vm for the voltage drop across a load).



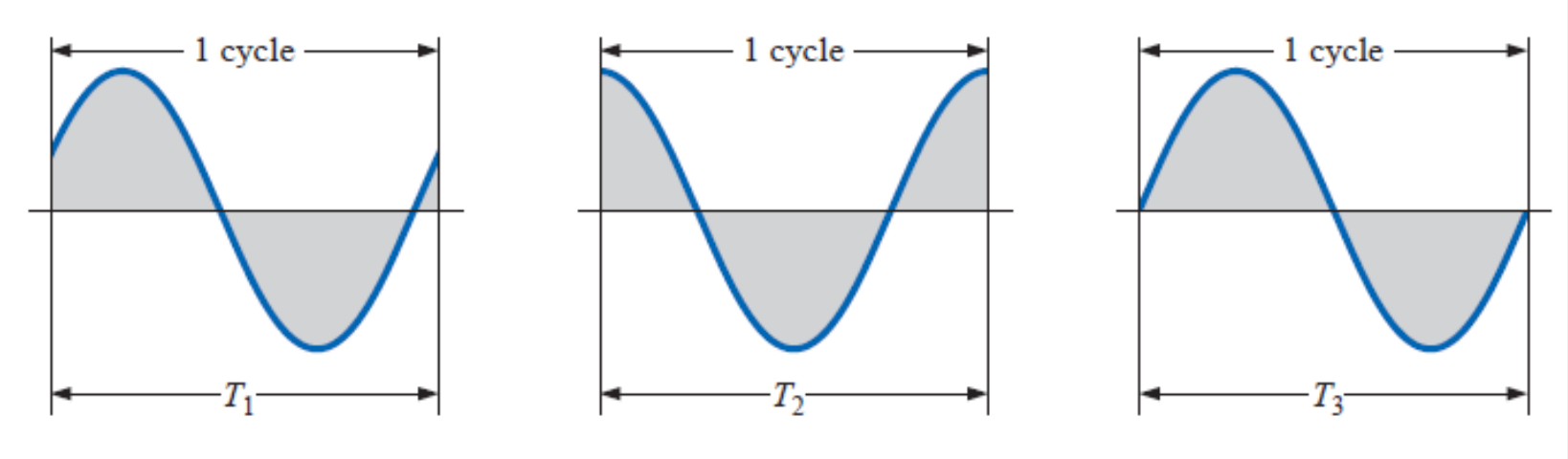
Peak-to-peak value: : The full voltage between positive and negative peaks of the waveform is known as Peak-to-peak value; denoted by Ep-p or Vp-p.



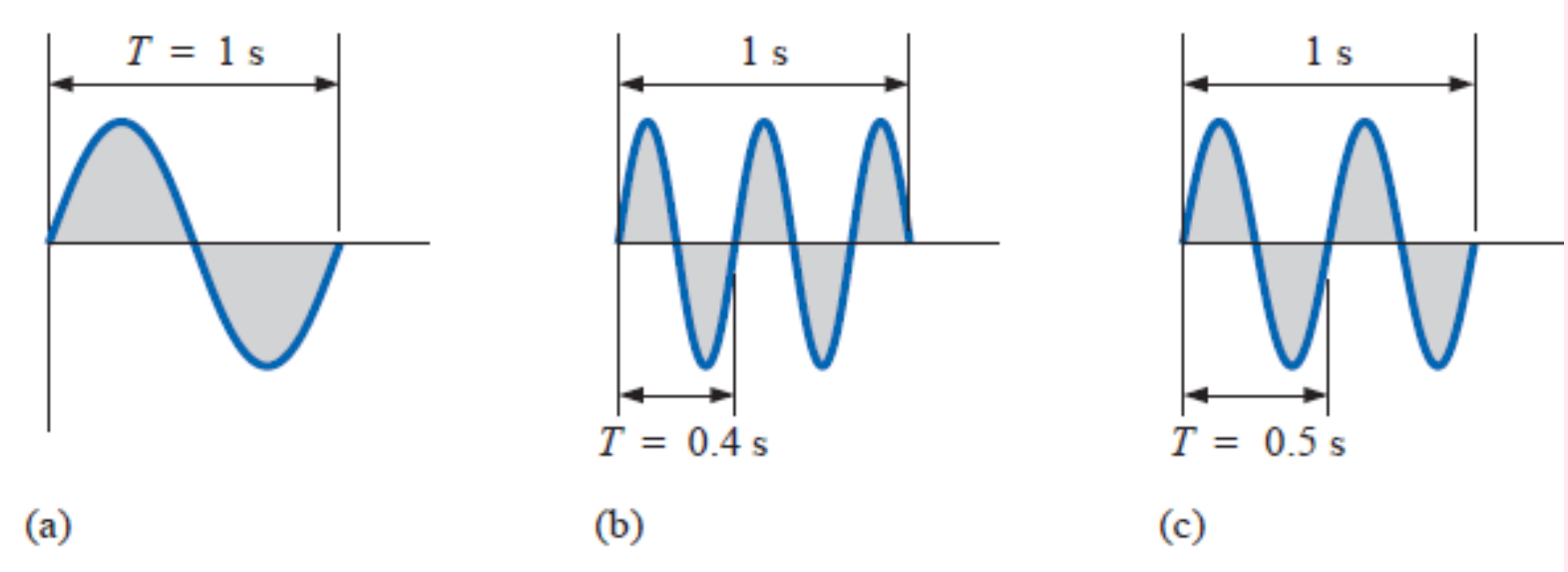
Periodic waveform: A waveform that continually repeats itself after the same time interval is known as periodic waveform. The waveform of following Fig. is a periodic waveform.



Period (T ): The time interval between successive repetitions of a periodic waveform (the period T1  T2  T3 in Following), is known as period.



Frequency ( f ): The number of cycles that occur in 1 s is known as frequency. The frequency of the waveform of Fig. (a) is 1 cycle per second, and for Fig.(b), 2.5 cycles per second. If a waveform of similar shape had a period of 0.5 s for Fig. (c),so the frequency would be 2 cycles per second.

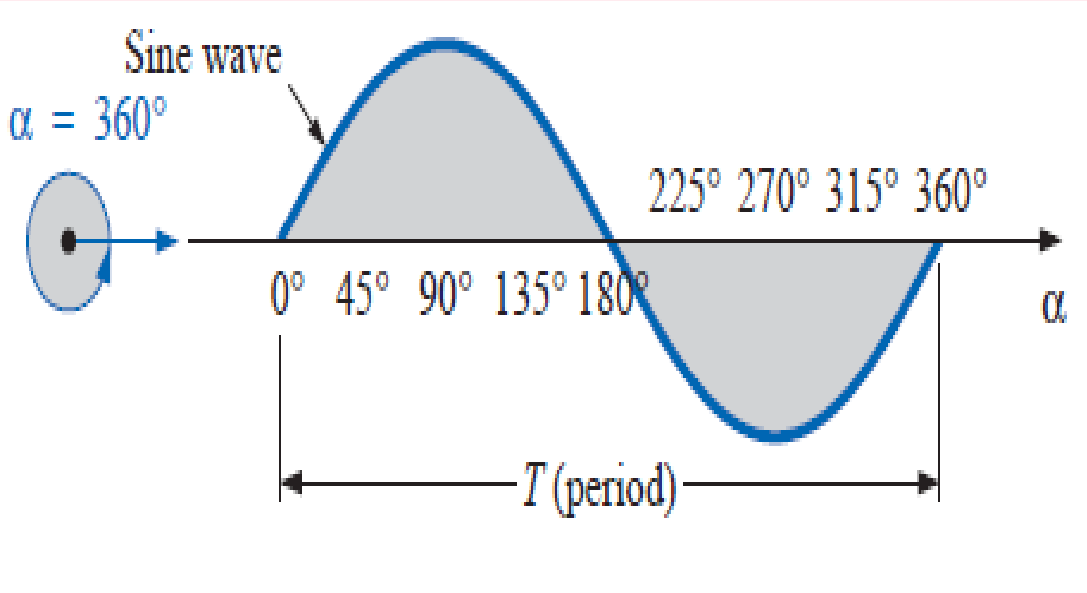


Angular velocity: The velocity with which the radius vector rotates about the center, called the angular velocity, can be determined from the following equation:

$$\text{Angular velocity} = \frac{\text{distance (degrees or radians)}}{\text{time (seconds)}}$$

$$\omega = \frac{\alpha}{t}$$

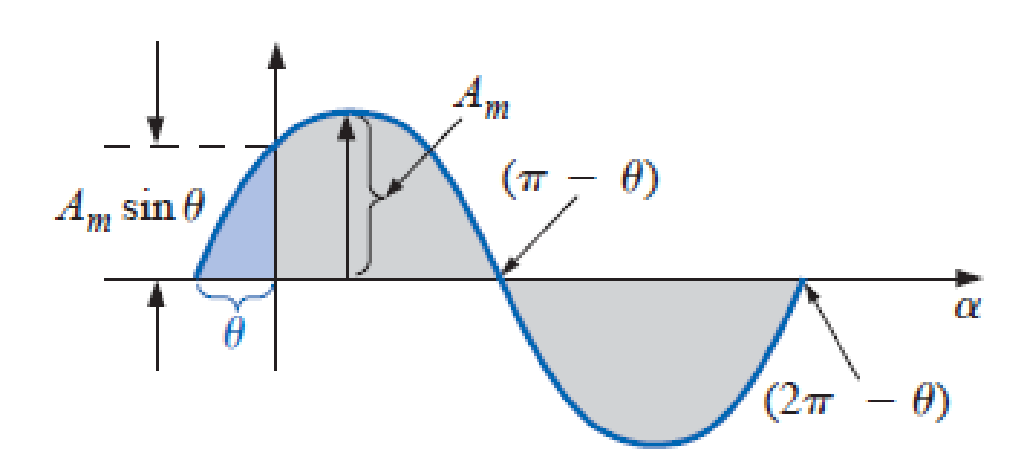
$$\omega = \frac{2\pi}{T}$$



# PHASE RELATIONS

If the waveform passes through the horizontal axis with a positive going (increasing with time) slope before 0°, as shown in given Fig., the expression is

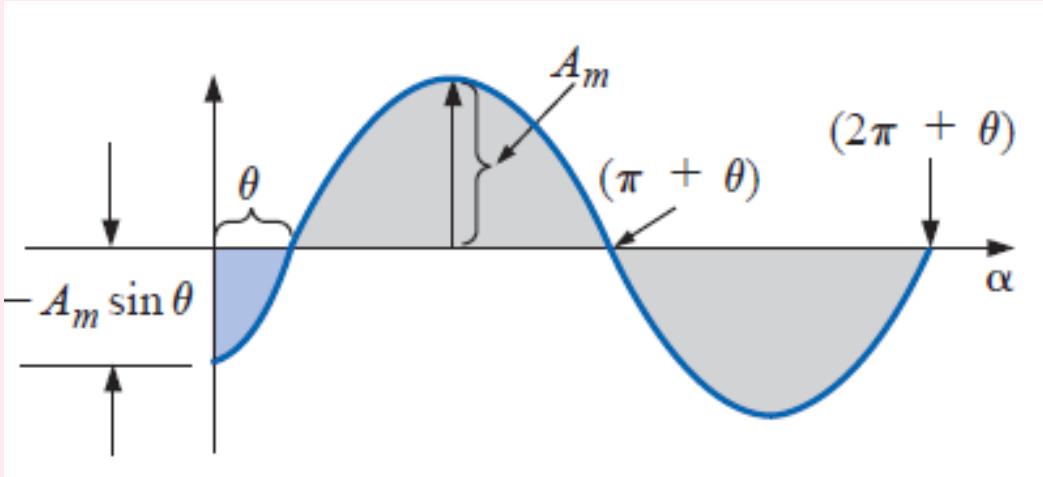
$$A_m \sin(\omega t + \theta)$$



# PHASE RELATIONS

If the waveform passes through the horizontal axis with a positive-going slope after 0°, as shown in given Fig., the expression is

$$A_m \sin(\omega t - \theta)$$



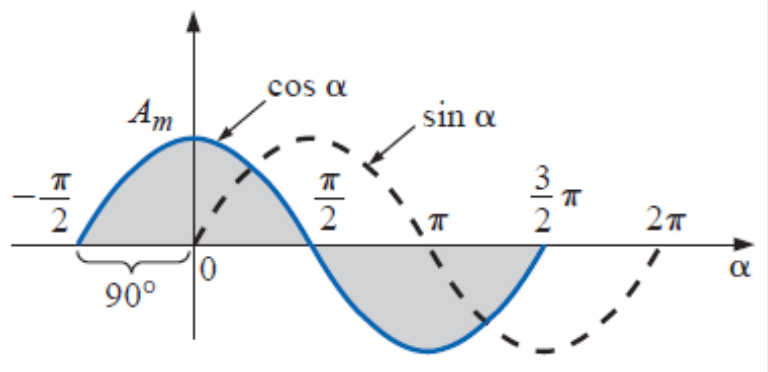
# PHASE RELATIONS

If the waveform crosses the horizontal axis with a positive-going slope 90° (π/2), as shown in following Fig., it is called a cosine wave; that is,

$$\sin(\omega t + 90°) = \sin\left(\omega t + \frac{\pi}{2}\right) = \cos \omega t$$

or

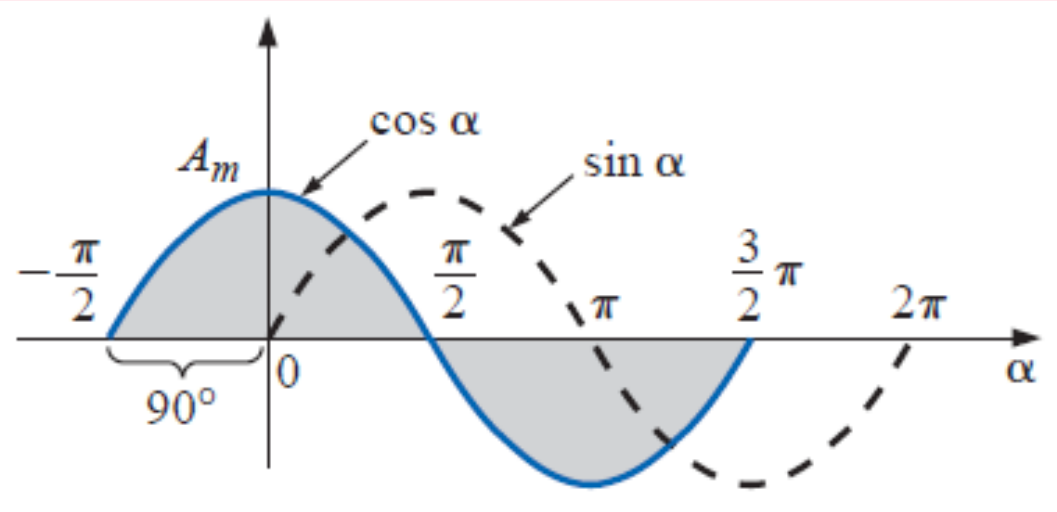
$$\sin \omega t = \cos(\omega t - 90°) = \cos\left(\omega t - \frac{\pi}{2}\right)$$

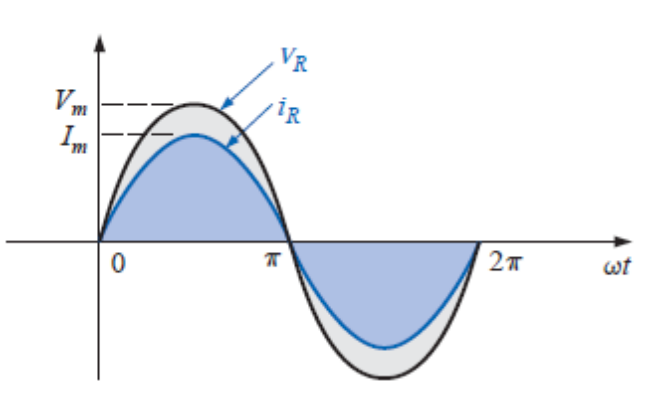


# LEAD AND LAG

In following Fig., the cosine curve is said to lead the sine curve by 90°, and the sine curve is said to lag the cosine curve by 90°.

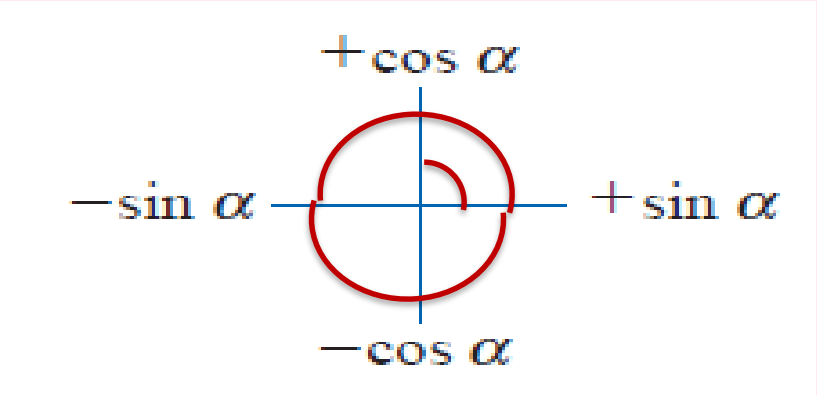
- In language commonly applied, the waveforms are out of phase by 90°.
- If both waveforms cross the axis at the same point with the same slope, they are in phase.



# LEAD AND LAG

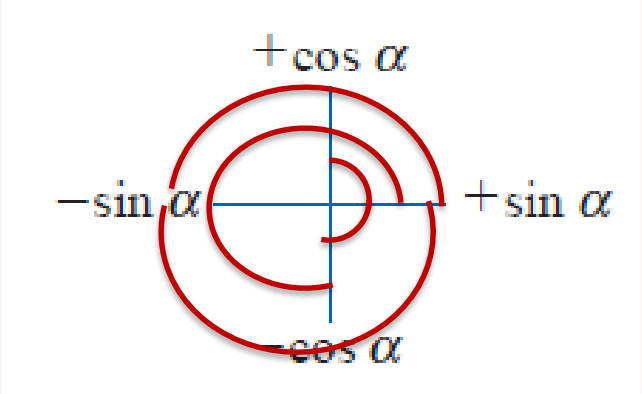
The geometric relationship between various forms of the sine and cosine functions can be derived from following Fig.




Starting at the sin**α** position, we find that cos**α** is an additional 90° in the counter clockwise direction. Therefore, cos**α**= sin(**α**+ 90°).

For sin**α** we must travel 180° in the counterclockwise (or clockwise) direction so that -sin**α**= sin(**α**± 180°)

# LEAD AND LAG




$$\cos\alpha = \sin(\alpha + 90°)$$
$$\sin\alpha = \cos(\alpha - 90°)$$
$$-\sin\alpha = \sin(\alpha \pm 180°)$$
$$-\cos\alpha = \sin(\alpha + 270°) = \sin(\alpha - 90°)$$
etc.

If a sinusoidal expression should appear as

$$e = -E_m \sin\omega t \longrightarrow e = E_m(-\sin\omega t)$$

$$-\sin\omega t = \sin(\omega t \pm 180°)$$

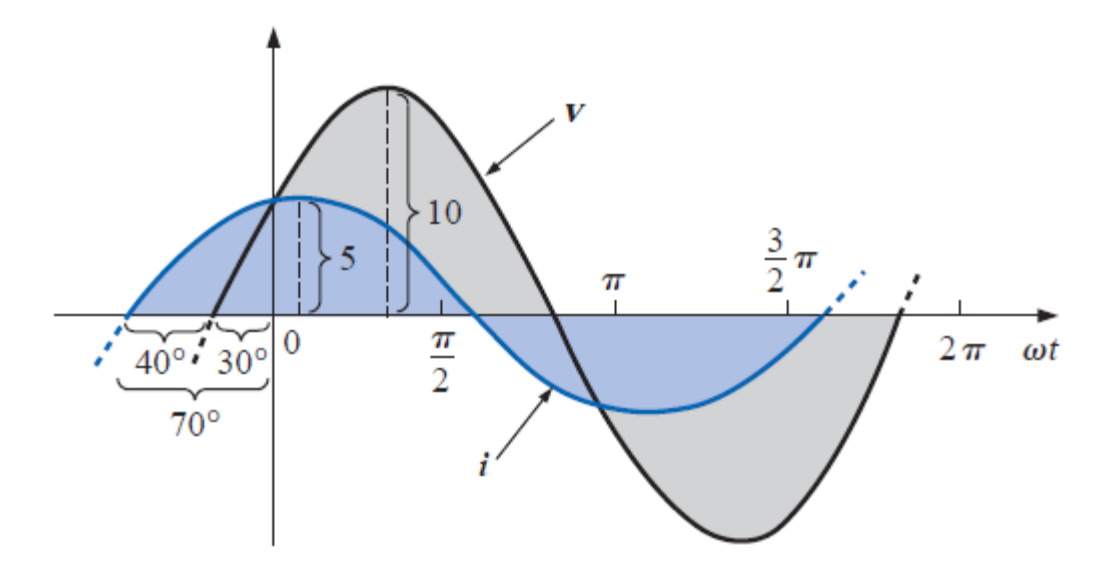
$$e = E_m \sin(\omega t \pm 180°)$$

Example 1: What is the phase relationship between the sinusoidal waveforms of the following sets?

a. $v = 10 \sin(\omega t + 30°)$
$i = 5 \sin(\omega t + 70°)$

Solutions:
v lags i by 40°or i leads v by 40°

30° -70° =-40°
v lags i by 40°

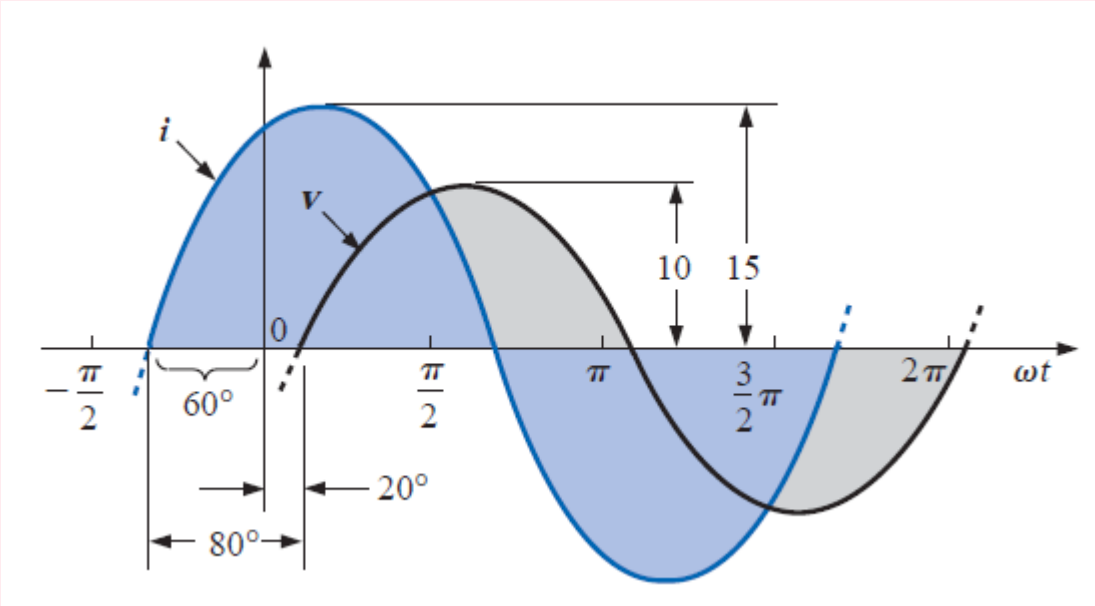


Example 2: What is the phase relationship between the sinusoidal waveforms of the following sets?

b. $i = 15 \sin(\omega t + 60°)$
$v = 10 \sin(\omega t - 20°)$

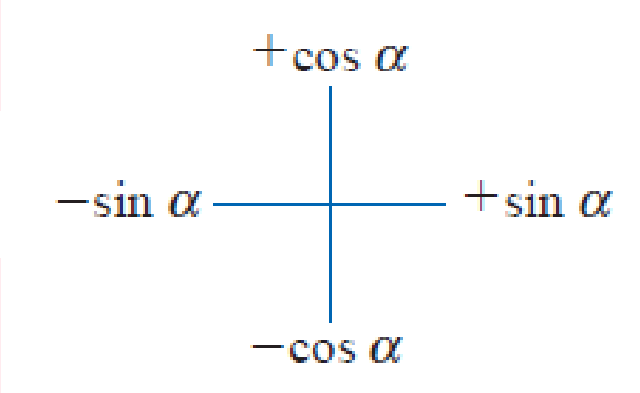
Solutions:
i leads v by 80°, or v lags i by 80°.

60° -(-20°)=80°
i leads v by 80°



Example 3: What is the phase relationship between the sinusoidal waveforms of the following sets?

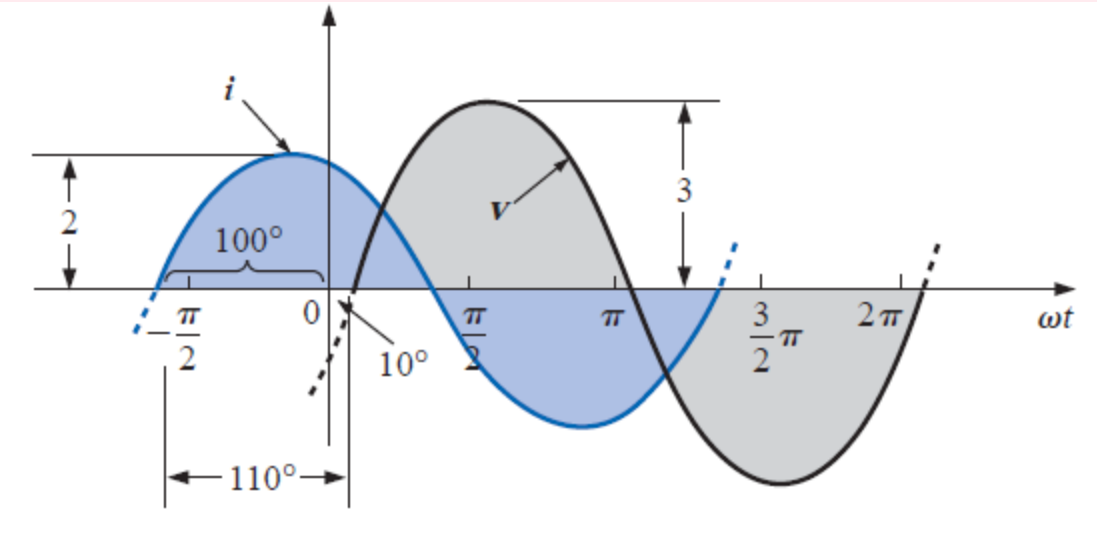
c. $i = 2\cos(\omega t + 10°)$
$v = 3\sin(\omega t - 10°)$




Solutions:

$i = 2\cos(\omega t + 10°) = 2\sin(\omega t +$
$= 2\sin(\omega t +$

i leads v by 110°, or v lags i by 110°.




100° -(-10°)=110°
i leads v by 110°

Example 4: What is the phase relationship between the sinusoidal waveforms of the following sets?

d. $i = -\sin(\omega t + 30°)$
$v = 2\sin(\omega t + 10°)$

Solutions:

-150° -10°=-160°
i lags v by 160°

210° -10°=200°
i leads v by 200°

$$-\sin(\omega t + 30°) = \sin(\omega t + 30° - 180°)$$
$$= \sin(\omega t - 150°)$$

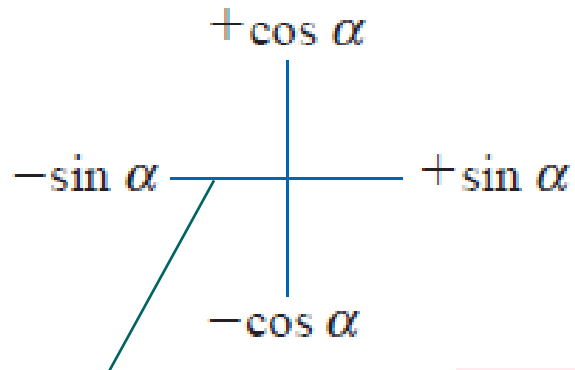
Note

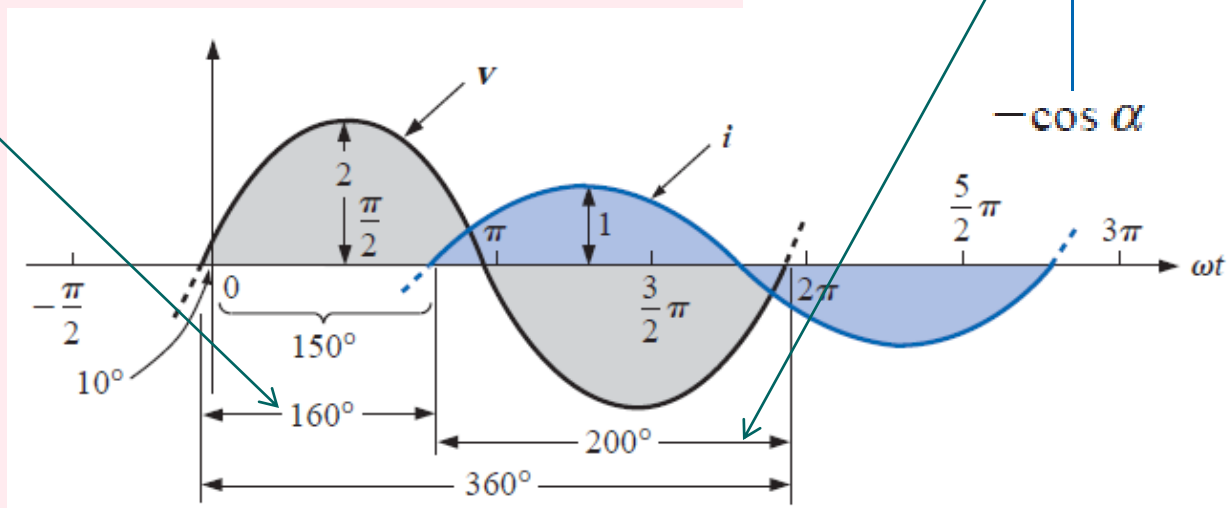
$$-\sin(\omega t + 30°) = \sin(\omega t + 30° + 180°)$$
$$= \sin(\omega t + 210°)$$

Note

i lea
by 2

i lags v by 160° or v leads i by160°



# Assignment

Assignment 1 : Find the phase angle between two equations and draw the wave shape

$$i_1 = -4\sin(377t + 25^\circ) \quad \text{and} \quad i_2 = 5\cos(377t - 40^\circ)$$

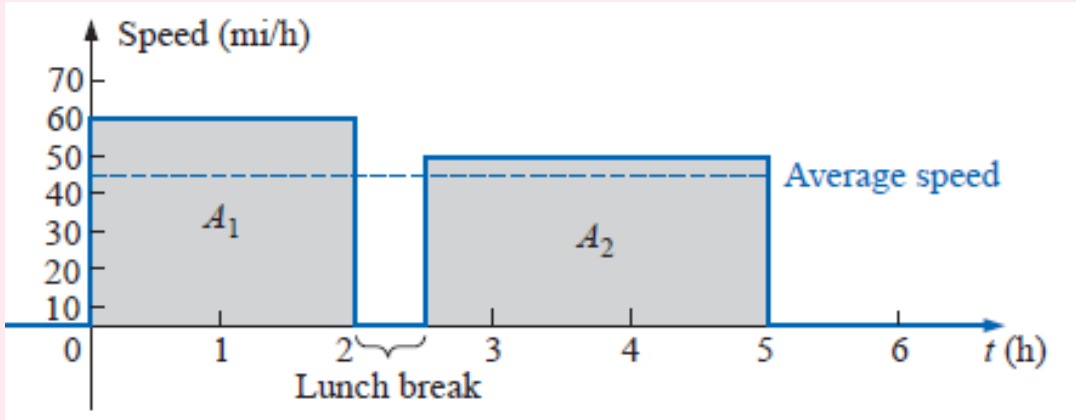
Assignment 2: Find the phase angle between two equations and draw the wave shape

$$v = -4\sin(30t + 50^\circ)$$

$$i = 4\sin(10t + 10^\circ)$$

# AVERAGE VALUE

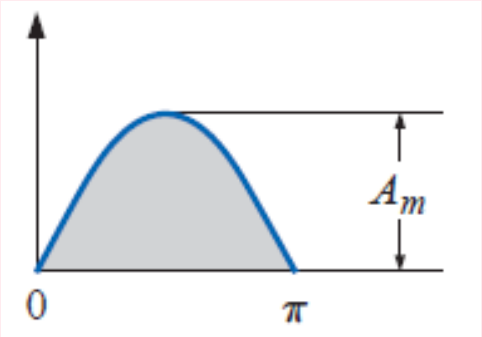
$$G \text{ (average value)} = \frac{\text{algebraic sum of areas}}{\text{length of curve}}$$

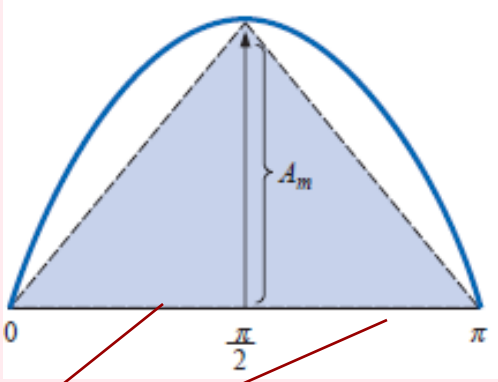



$$\text{Average speed} = \frac{\text{area under curve}}{\text{length of curve}}$$

$$\begin{aligned}
\text{Average speed} &= \frac{A_1 + A_2}{5 \text{ h}} \\
&= \frac{(60 \text{ mi/h})(2 \text{ h}) + (50 \text{ mi/h})(2.5 \text{ h})}{5 \text{ h}} \\
&= \frac{225}{5} \text{ mi/h} \\
&= \mathbf{45 \text{ mi/h}}
\end{aligned}$$

# AVERAGE VALUE of half sine wave

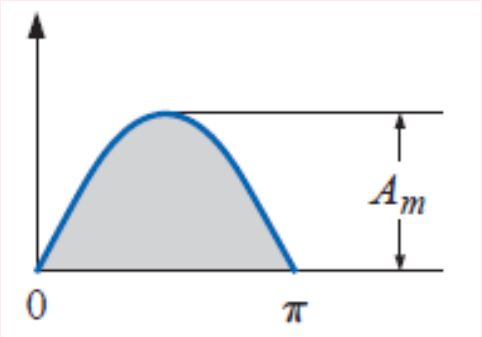


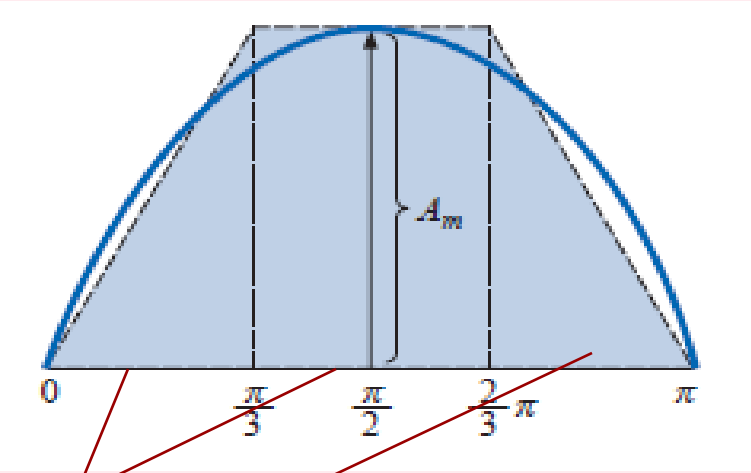




$$\text{Area shaded} = 2\left(\frac{1}{2}bh\right) = 2\left[\left(\frac{1}{2}\right)\overbrace{\left(\frac{\pi}{2}\right)}^{b}\overbrace{(A_m)}^{h}\right] = \frac{\pi}{2}A_m$$

$$\cong 1.58A_m$$

# AVERAGE VALUE of half sine wave




$$\text{Area} = A_m\frac{\pi}{3} + 2\left(\frac{1}{2}bh\right) = A_m\frac{\pi}{3} + \frac{\pi}{3}A_m = \frac{2}{3}\pi A_m$$

$$= 2.094A_m$$

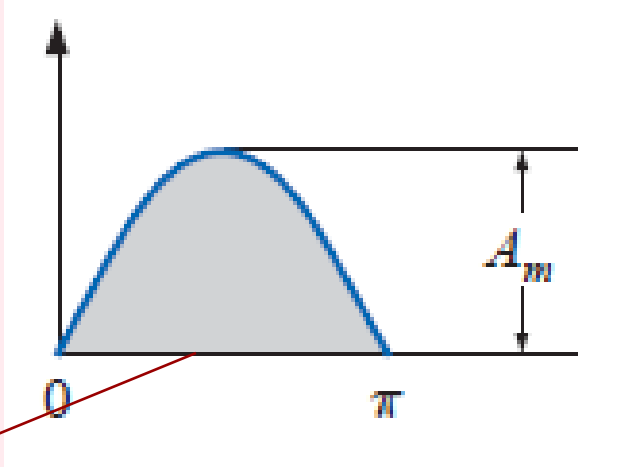
# AVERAGE VALUE of half sine wave

Prove that the area of the positive (or negative) pulse of a sine wave is 2Am.

$$\text{Area} = \int_0^{\pi} A_m \sin \alpha \, d\alpha$$

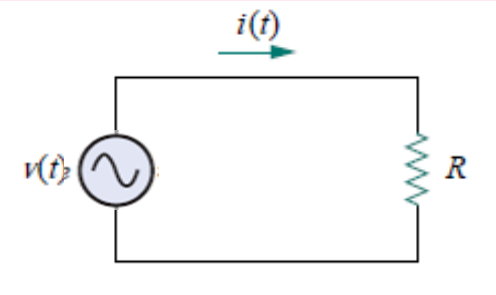
$$\begin{aligned}\text{Area} &= A_m[-\cos \alpha]_0^{\pi} \\ &= -A_m(\cos \pi - \cos 0°) \\ &= -A_m[-1 - (+1)] = -A_m(-2)\end{aligned}$$

$$\boxed{\text{Area} = 2A_m}$$



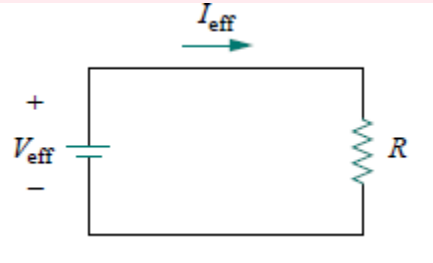
# EFFECTIVE (root-mean-square[rms]) VALUES

The effective value of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.




The average power absorbed by the resistor in the ac circuit is

$$P = \frac{1}{T}\int_0^T i^2 R\,dt = \frac{R}{T}\int_0^T i^2\,dt$$




while the power absorbed by the resistor in the dc circuit is

$$P = I_{eff}^2 R$$

# EFFECTIVE (root-mean-square[rms]) VALUES

$$I_{\text{eff}} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt}$$

$$I_{\text{eff}} = \sqrt{\frac{\text{area}\,(i^2(t))}{T}}$$

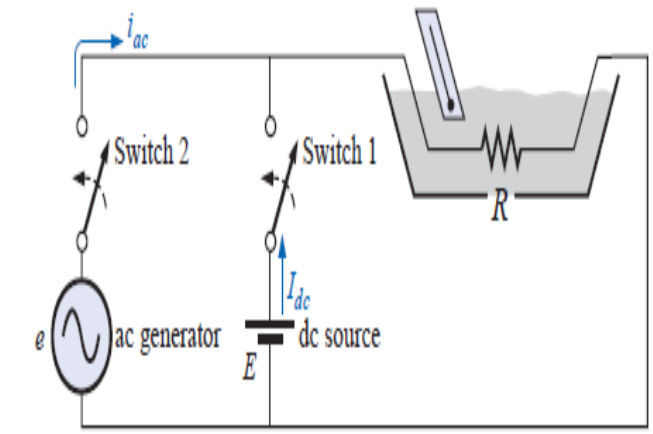
# EFFECTIVE (root-mean-square[rms]) VALUES

The power delivered by the ac supply at any instant of time is

$$P_{ac} = (i_{ac})^2 R = (I_m \sin \omega t)^2 R = (I_m^2 \sin^2 \omega t) R$$

but

$$\sin^2 \omega t = \frac{1}{2}(1 - \cos 2\omega t) \qquad \text{(trigonometric identity)}$$




Therefore,

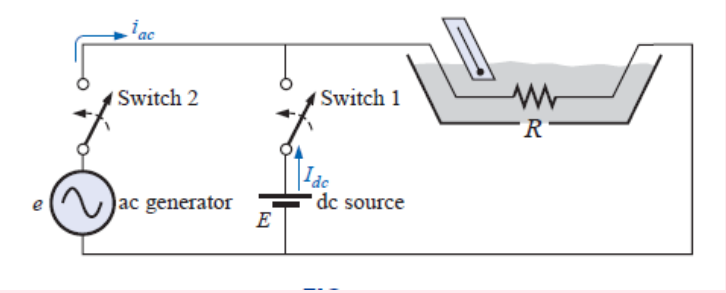
$$P_{ac} = I_m^2 \left[ \frac{1}{2}(1 - \cos 2\omega t) \right] R$$

and

$$\boxed{P_{ac} = \frac{I_m^2 R}{2} - \frac{I_m^2 R}{2} \cos 2\omega t}$$

# EFFECTIVE (root-mean-square[rms]) VALUES

$$P_{ac} = \frac{I_m^2 R}{2} - \frac{I_m^2 R}{2} \cos 2\omega t$$




$$P_{av(ac)} = \frac{I_m^2 R}{2}$$

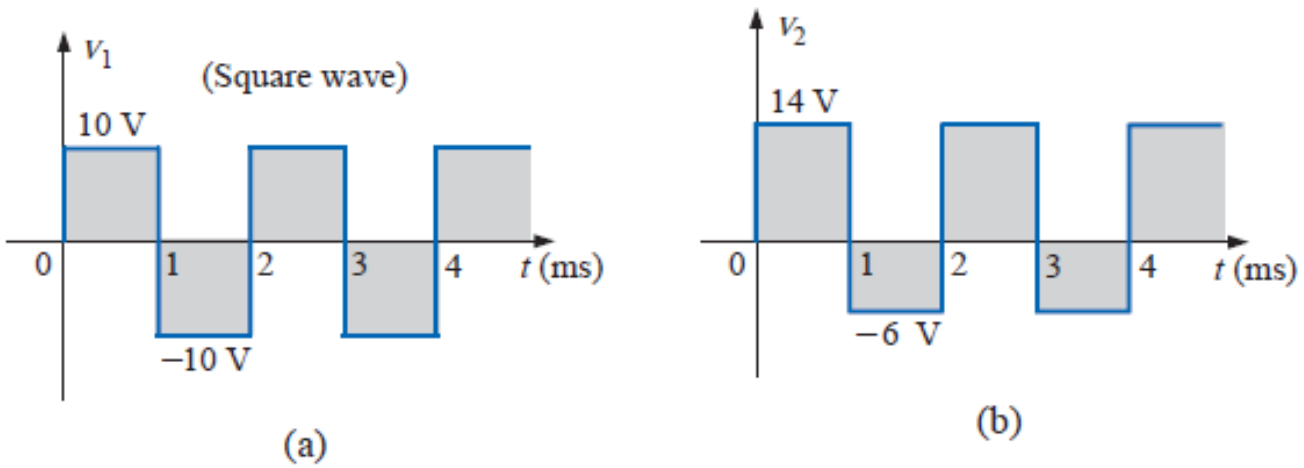
$$P_{av(ac)} = P_{dc}$$

$$\frac{I_m^2 R}{2} = I_{dc}^2 R \quad \text{and} \quad I_m = \sqrt{2} I_{dc}$$

$$I_{dc} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

the equivalent dc value of a sinusoidal current or voltage is $1/\sqrt{2}$ or 0.707 of its maximum value.

Determine the average value of the waveforms of

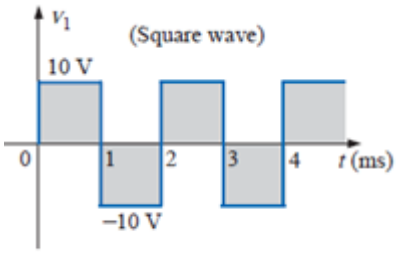


## Solutions:

a. By inspection, the area above the axis equals the area below over one cycle, resulting in an average value of zero volts. Using Eq. (13.26):

$$G = \frac{(10\text{ V})(1\text{ ms}) - (10\text{ V})(1\text{ ms})}{2\text{ ms}}$$

$$= \frac{0}{2\text{ ms}} = \mathbf{0\ V}$$

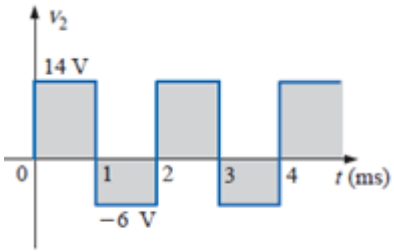


(a)

b. Using Eq. (13.26):

$$G = \frac{(14\text{ V})(1\text{ ms}) - (6\text{ V})(1\text{ ms})}{2\text{ ms}}$$

$$= \frac{14\text{ V} - 6\text{ V}}{2} = \frac{8\text{ V}}{2} = \mathbf{4\ V}$$



(b)

Calculate the rms value of the voltage of Fig. 13.57.

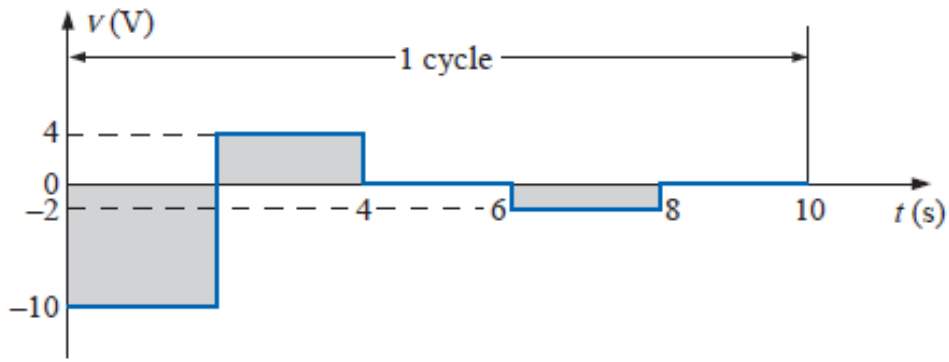



**FIG. 13.57**
*Example 13.22.*

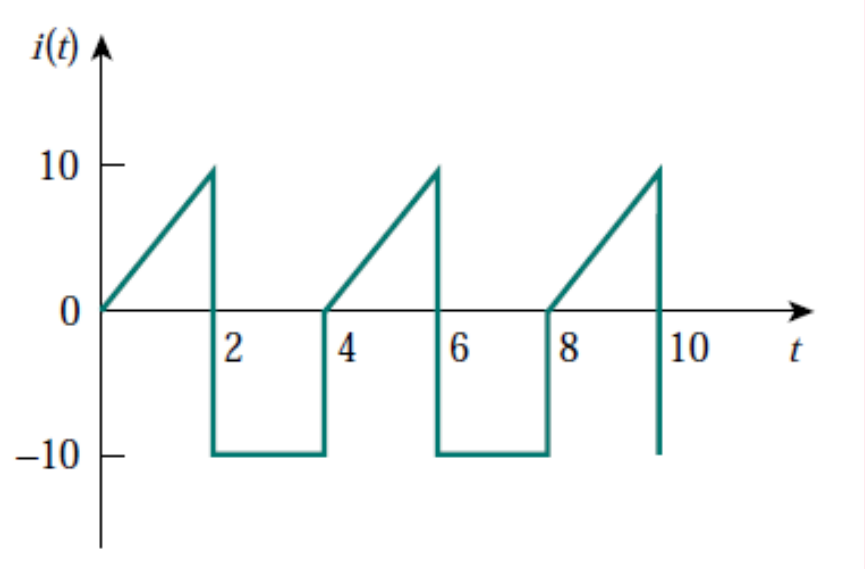
***Solution:***

$v^2$ (Fig. 13.58):

$$V_{rms} = \sqrt{\frac{(100)(2) + (16)(2) + (4)(2)}{10}} = \sqrt{\frac{240}{10}}$$

$$= \mathbf{4.899\ V}$$

Determine the rms value of the current waveform in Fig. 11.14. If the current is passed through a 2-Ω resistor, find the average power absorbed by the resistor.




y=mx

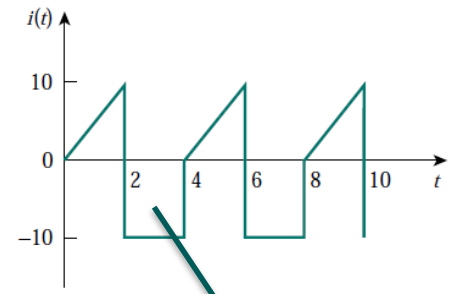
$$i(t) = mt$$

$$i(t) = mt = \tan\theta t = \frac{\Delta i}{\Delta t} t = \frac{i_2 - i_1}{t_2 - t_1} t = \frac{10-0}{2-0} t = 5t$$

**Solution:**

The period of the waveform is $T = 4$. Over a period, we can write the current waveform as

$$i(t) = \begin{cases} 5t, & 0 < t < 2 \\ -10, & 2 < t < 4 \end{cases}$$




The rms value is

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt} = \sqrt{\frac{1}{4}\left[\int_0^2 (5t)^2\,dt + \int_2^4 (-10)^2\,dt\right]}$$

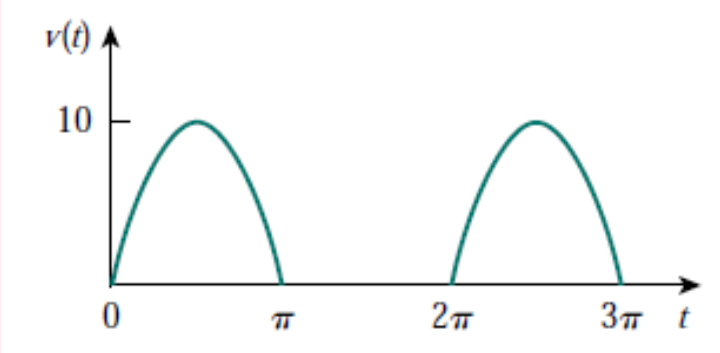
$$(-10)^2 \times 2 = 200$$

$$= \sqrt{\frac{1}{4}\left[25\frac{t^3}{3}\bigg|_0^2 + 100t\bigg|_2^4\right]} = \sqrt{\frac{1}{4}\left(\frac{200}{3} + 200\right)} = 8.165 \text{ A}$$

The power absorbed by a 2-Ω resistor is

$$P = I_{\text{rms}}^2 R = (8.165)^2(2) = 133.3 \text{ W}$$

The waveform shown in Fig. 11.16 is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a 10-Ω resistor.



**Solution:**

The period of the voltage waveform is $T = 2\pi$, and

$$v(t) = \begin{cases} 10\sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

The rms value is obtained as

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v^2(t)\,dt = \frac{1}{2\pi}\left[\int_0^{\pi}(10\sin t)^2\,dt + \int_{\pi}^{2\pi} 0^2\,dt\right]$$

But $\sin^2 t = \frac{1}{2}(1 - \cos 2t)$. Hence

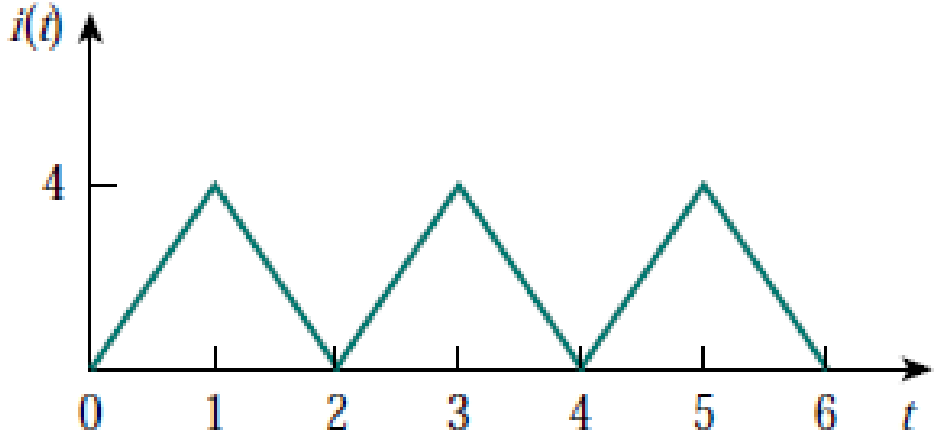
$$V_{\text{rms}}^2 = \frac{1}{2\pi}\int_0^{\pi}\frac{100}{2}(1 - \cos 2t)\,dt = \frac{50}{2\pi}\left(t - \frac{\sin 2t}{2}\right)\Bigg|_0^{\pi}$$

$$= \frac{50}{2\pi}\left(\pi - \frac{1}{2}\sin 2\pi - 0\right) = 25, \qquad V_{\text{rms}} = 5\text{ V}$$

The average power absorbed is

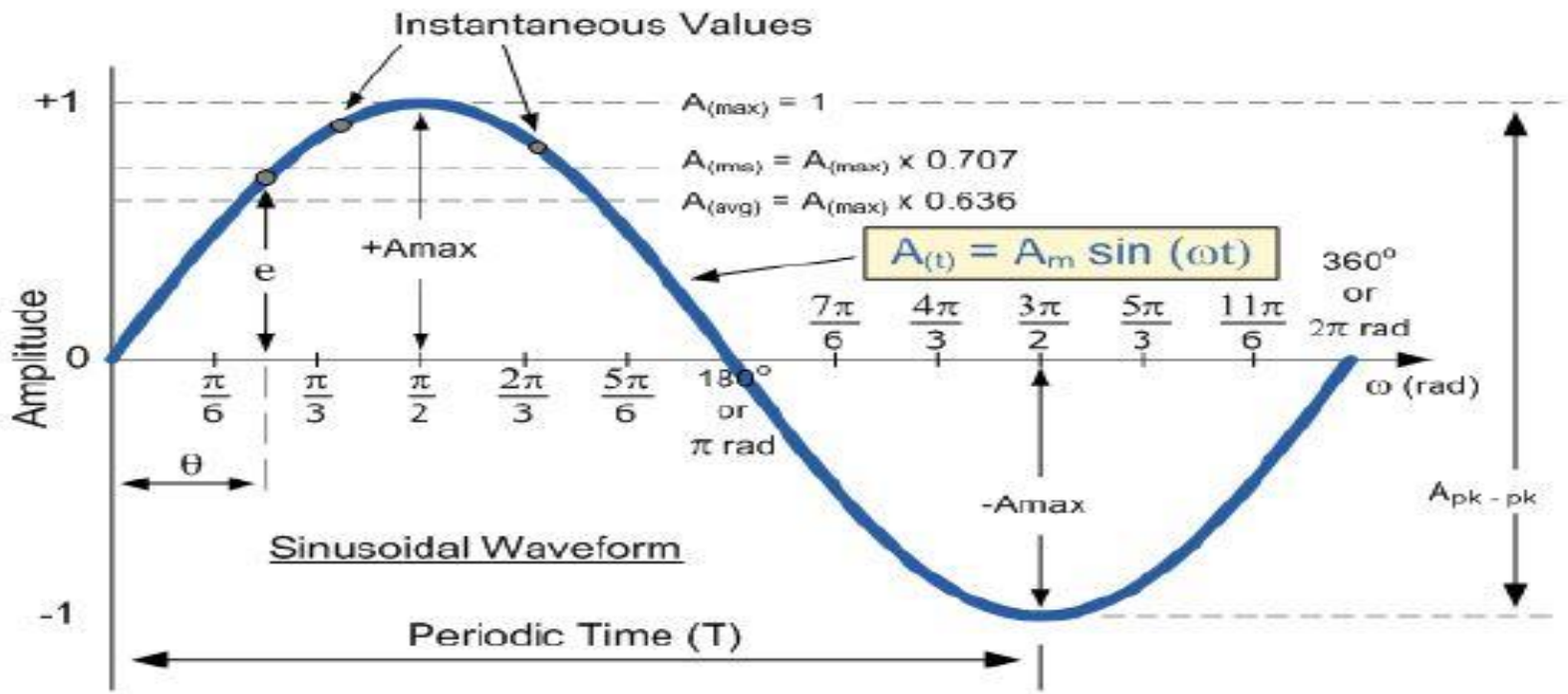
$$P = \frac{V_{\text{rms}}^2}{R} = \frac{5^2}{10} = 2.5\text{ W}$$

Find the rms value of the current waveform of Fig. 11.15. If the current flows through a 9-Ω resistor, calculate the average power absorbed by the resistor.




**Answer:** 2.309 A, 48 W.

# SINUSOIDAL WAVEFORM



AVERAGE VALUE : The average of all instantaneous values during one alternation

average value ($V_{avg}$) = 0.637Em

FORM FACTOR : The ratio between rms value and average value

$$\frac{V_{rms}}{V_{avg}} = 1.11$$

PEAK FACTOR : peak value / 0.707 peak value = 1.414

# QUESTION

- Equation for an alternating current is :

  $I = 70.71 \sin 520t$

Determine :

i) Peak current value

ii) Rms current value

iii) Average current value

iv) frequency

# ANSWER

i) Peak value ($I_p$) = 70.71 A

ii) Rms value ($I_{rms}$) = $0.707 I_p$

$= 0.707 \times 70.71 = 50$ A

iii) Average value ($I_{avg}$) = $0.637 I_p$

$= 0.637 \times 70.71$

$= 45$ A

iv) $\omega = 2\pi f$ → f=82.76 Hz

# QUESTION

A sinusoidal waveform is defined as:
$V_m = 169.8 \sin(377t)$ volts. Calculate the RMS voltage of the waveform, its frequency and the instantaneous value of the voltage after a time of 6mS.

# ANSWER

$$V_{(t)} = V_m \sin(\omega t)$$

Then comparing this to our given expression for a sinusoidal waveform above of $V_m$ = 169.8 sin(377t) will give us the peak voltage value of 169.8 volts for the waveform.

$$V_{(rms)} = 0.707 \times \text{maximum peak value}$$

$$V_{(rms)} = 0.707 \times 169.8 = 120 \text{volts}$$

The angular velocity (ω) is given as 377 rad/s. Then $2\pi f = 377$. So the frequency of the waveform is calculated as:

$$\text{Frequency}, (f) = \frac{377}{2\pi} = 60\text{Hz}$$

The instantaneous voltage $V_i$ value after a time of 6mS is given as:

$$V_{(i)} = 169.8 \sin(377 \times 6\text{mS})$$

$$V_{(i)} = 169.8 \sin(2.262)$$

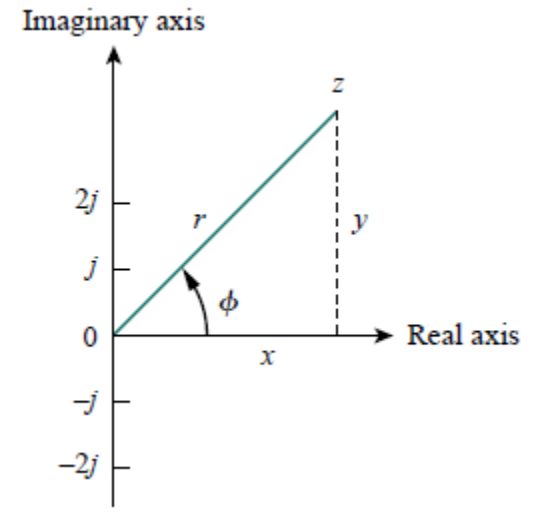
$$\therefore V_{(i)} = 130.8 \text{ volts}$$

# COMPLEX NUMBER

A complex number z can be written in rectangular form as
$z = x + jy$
where $j = \sqrt{-1}$; x is the real part of z; y is the imaginary part of z.

The complex number z can also be written in polar or exponential form as

$$z = r\angle\phi = re^{j\phi}$$




Z can be represented in three ways

| | |
|---|---|
| $z = x + jy$ | Rectangular form |
| $z = r\angle\phi$ | Polar form |
| $z = re^{j\phi}$ | Exponential form |

# COMPLEX NUMBER

x and y, we can get r and φ as.

$$r = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}$$

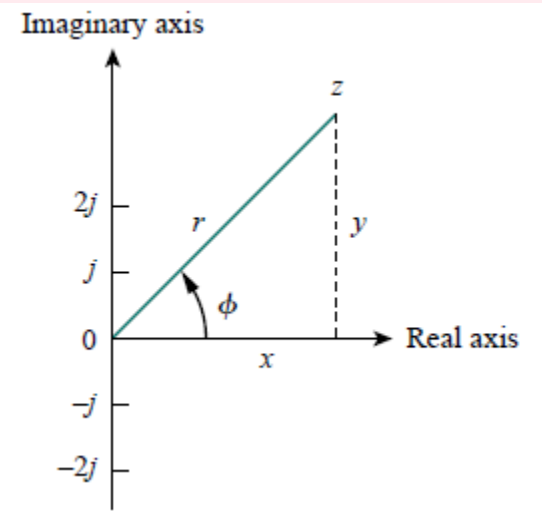
On the other hand, if we know r and φ, we can obtain x and y as

x = r cos φ, y = r sin φ

z = x + jy = r<φ = r(cos φ + j sin φ)

Consider the complex numbers

$$z = x + jy = r\angle\phi, \qquad z_1 = x_1 + jy_1 = r_1\angle\phi_1$$
$$z_2 = x_2 + jy_2 = r_2\angle\phi_2$$



# COMPLEX NUMBER

**Addition:**

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

**Subtraction:**

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

**Multiplication:**

$$z_1 z_2 = r_1 r_2 \underline{/\phi_1 + \phi_2}$$

**Division:**

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \underline{/\phi_1 - \phi_2}$$

**Reciprocal:**

$$\frac{1}{z} = \frac{1}{r} \underline{/-\phi}$$

**Square Root:**

$$\sqrt{z} = \sqrt{r} \underline{/\phi/2}$$

# COMPLEX NUMBER

**Complex Conjugate:**

$$z^* = x - jy = r\angle -\phi = re^{-j\phi}$$

The idea of phasor representation is based on Euler's identity

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

$$\cos\phi = \mathrm{Re}(e^{j\phi})$$
$$\sin\phi = \mathrm{Im}(e^{j\phi})$$

# COMPLEX NUMBER

## Evaluate these complex numbers

(a) $(40\angle 50^\circ + 20\angle -30^\circ)^{1/2}$

(a) Using polar to rectangular transformation,

$$40\angle 50^\circ = 40(\cos 50^\circ + j\sin 50^\circ) = 25.71 + j30.64$$

$$20\angle -30^\circ = 20[\cos(-30^\circ) + j\sin(-30^\circ)] = 17.32 - j10$$

Adding them up gives

$$40\angle 50^\circ + 20\angle -30^\circ = 43.03 + j20.64 = 47.72\angle 25.63^\circ$$

Taking the square root of this,

$$(40\angle 50^\circ + 20\angle -30^\circ)^{1/2} = 6.91\angle 12.81^\circ$$

# COMPLEX NUMBER

## Evaluate these complex numbers

(b) $\dfrac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}$

(b) Using polar-rectangular transformation, addition, multiplication, and division,

$$\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*} = \frac{8.66 - j5 + (3 - j4)}{(2 + j4)(3 + j5)}$$

$$= \frac{11.66 - j9}{-14 + j22} = \frac{14.73\angle -37.66^\circ}{26.08\angle 122.47^\circ}$$

$$= 0.565\angle -160.31^\circ$$

# COMPLEX NUMBER

Evaluate the following complex numbers:

(a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$

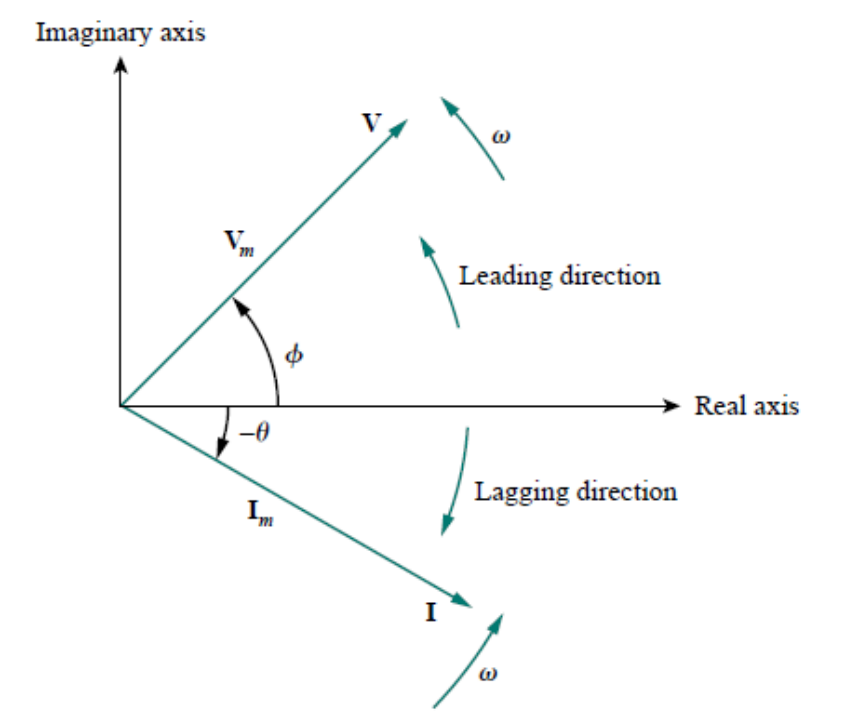
(b) $\dfrac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ$

**Answer:** (a) $-15.5 - j13.67$, (b) $8.293 + j2.2$.

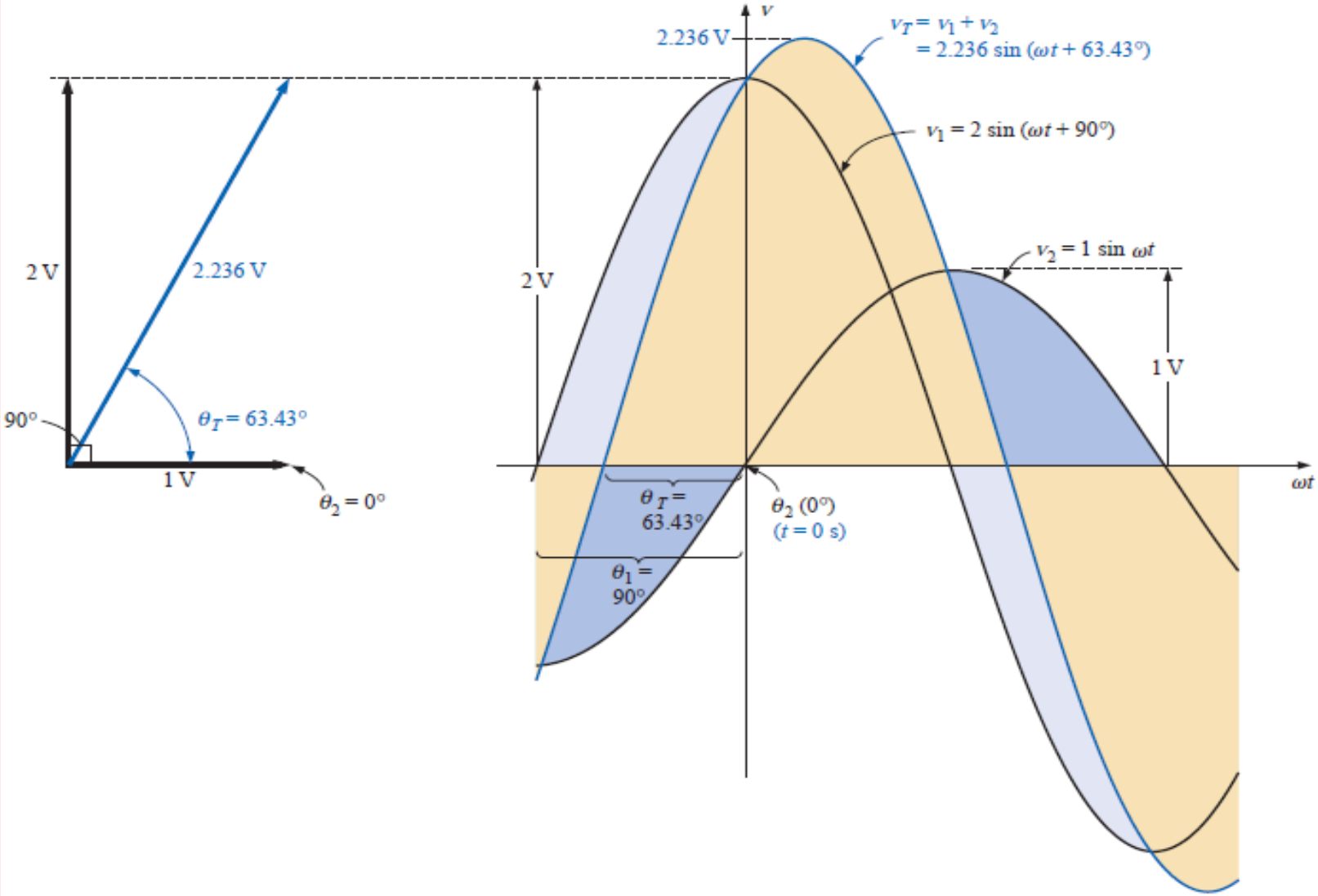
# PHASOR

This radius vector, having a constant magnitude (length) with one end fixed at the origin, is called a phasor when applied to electric circuits.

phasors V = Vm<φ and I = Im <−θ are graphically represented in following Fig. Such a graphical representation of phasors is known as a phasor diagram.

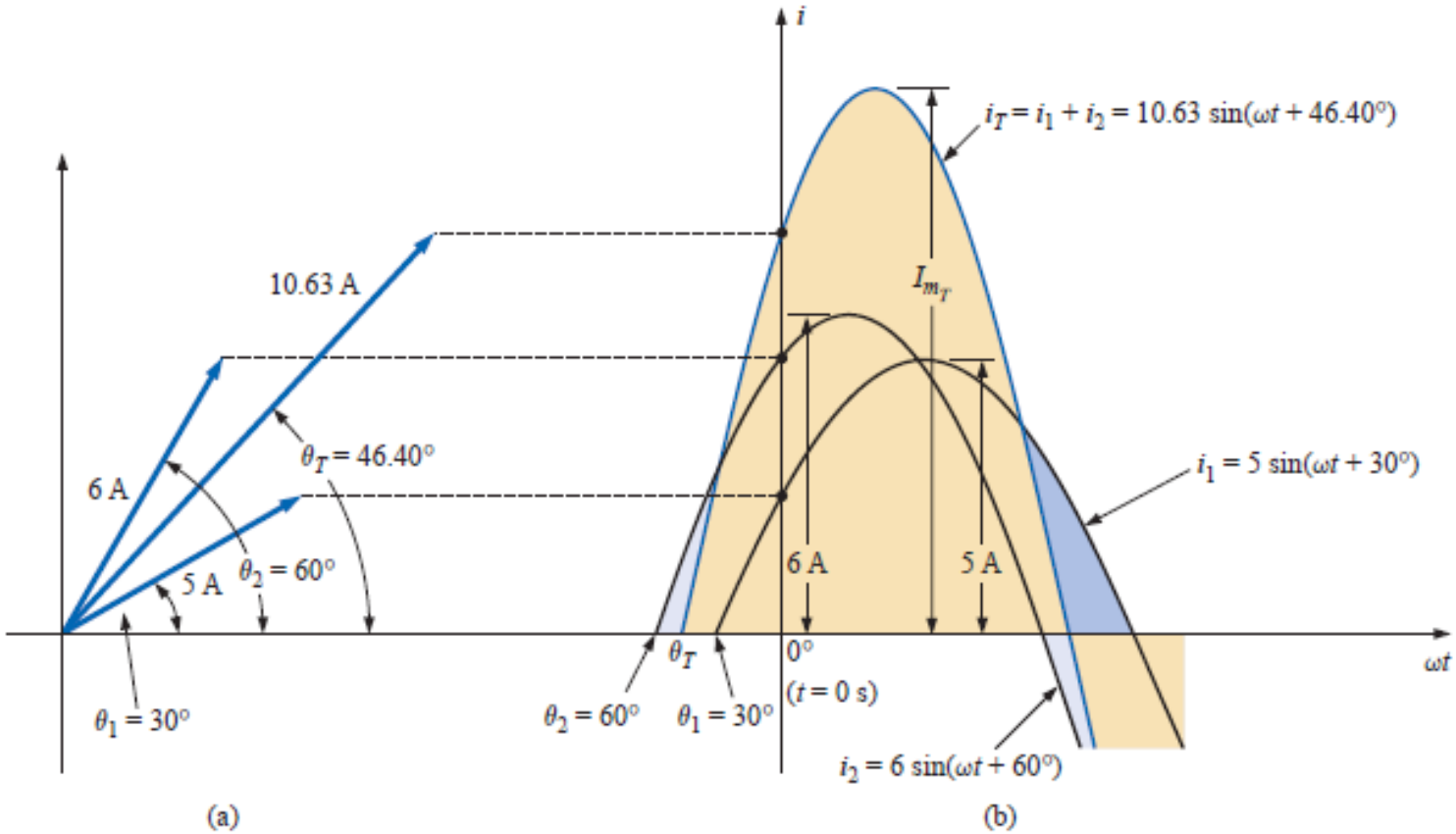


# PHASOR


2 V
2.236 V
90°
$\theta_T = 63.43°$
1 V
$\theta_2 = 0°$
v
2.236 V
$v_T = v_1 + v_2$
$= 2.236 \sin(\omega t + 63.43°)$
$v_1 = 2 \sin(\omega t + 90°)$
$v_2 = 1 \sin \omega t$
2 V
1 V
$\omega t$
$\theta_T =$
63.43°
$\theta_2$ (0°)
($t = 0$ s)
$\theta_1 =$
90°

# PHASOR

Phasor algebra for sinusoidal quantities is applicable only for waveforms having the same frequency.

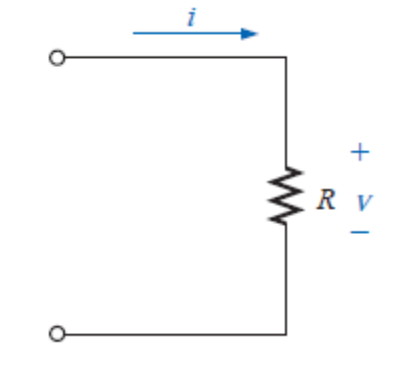


# RESPONSE OF BASIC R, L, AND C ELEMENTS TO A SINUSOIDAL VOLTAGE OR CURRENT

## RESISTOR

For power-line frequencies and frequencies up to a few hundred kilohertz, resistance is, for all practical purposes, unaffected by the frequency of the applied sinusoidal voltage or current. For this frequency region, the resistor R of Following Fig. can be treated as a constant, and Ohm's law can be applied as follows. For v= Vm sin $\omega t$

$$i = \frac{v}{R} = \frac{V_m \sin \omega t}{R} = \frac{V_m}{R} \sin \omega t = I_m \sin \omega t$$



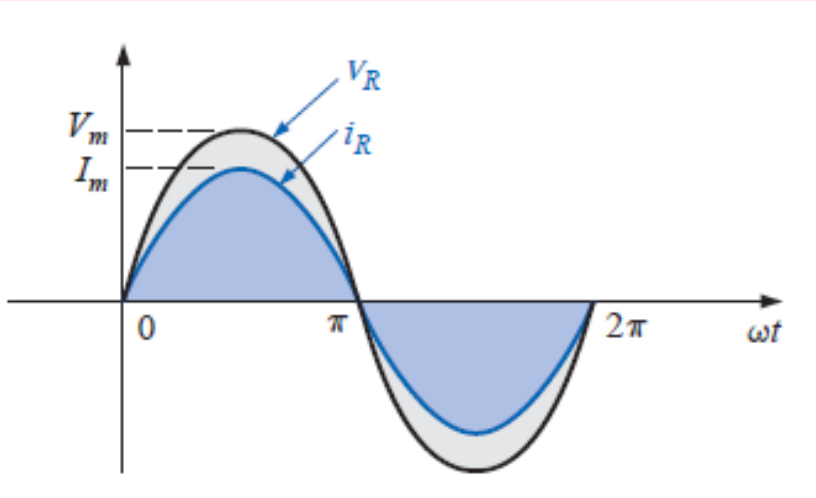

Where $I_m = \frac{V_m}{R}$

In addition, for a given i,

$$v = iR = (I_m \sin \omega t)R = I_m R \sin \omega t = V_m \sin \omega t$$

# RESISTOR

A plot of v and i in following Fig. reveals that




for a purely resistive element, the voltage across and the current through the element are in phase, with their peak values related by Ohm's law.

# INDUCTOR

The inductive voltage is directly related to the frequency and the inductance of the coil. For increasing values of f and L the magnitude of $v_L$ will increase.

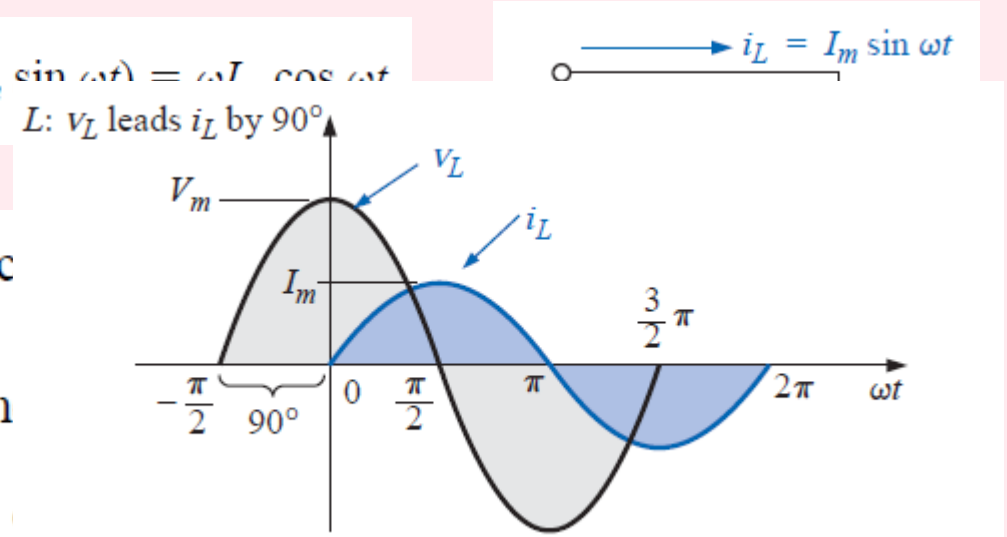
$$v_L = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{d}{dt}(I_m \sin \omega t) = \omega I_m \cos \omega t$$

Therefore, $v_L = L\dfrac{di_L}{dt} = L(\omega I_m c$

or $v_L = V_m \sin$

where $V_m =$




for an inductor, $V_L$ leads $i_L$ by 90°, or $i_L$ lags $V_L$ by 90°.

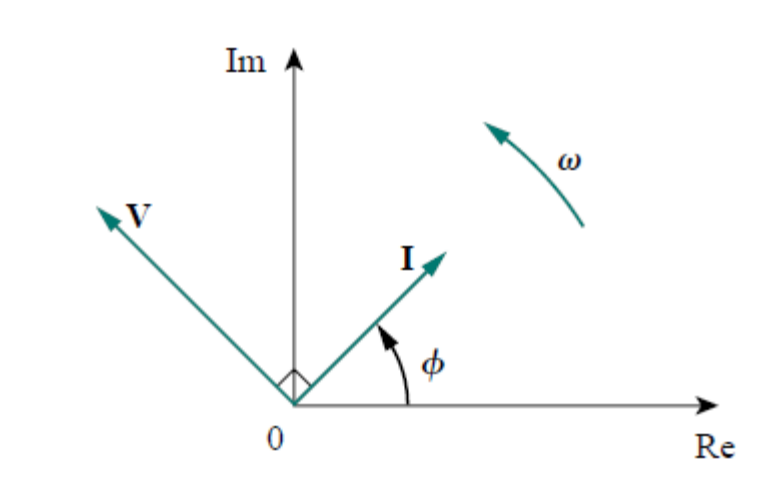
# INDUCTOR

If a phase angle is included in the sinusoidal expression for $i_L$, such as

$$i_L = I_m \sin(\omega t \pm \theta)$$

then

$$v_L = \omega L I_m \sin(\omega t \pm \theta + 90°)$$

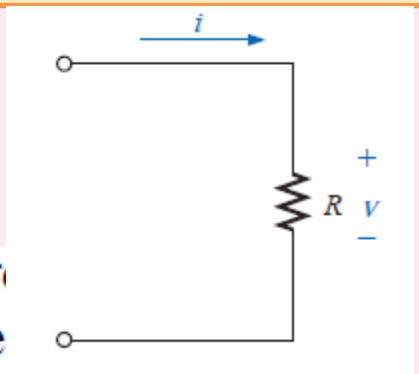



Phasor diagram for the inductor; I lags V.

# INDUCTOR

The opposition established by an inductor in a sinusoidal ac network can now be found by

$$\text{Opposition} = \frac{\text{cause}}{\text{effect}} = \frac{V_m}{I_m} = \frac{\omega L I_m}{I_m} = \omega L$$



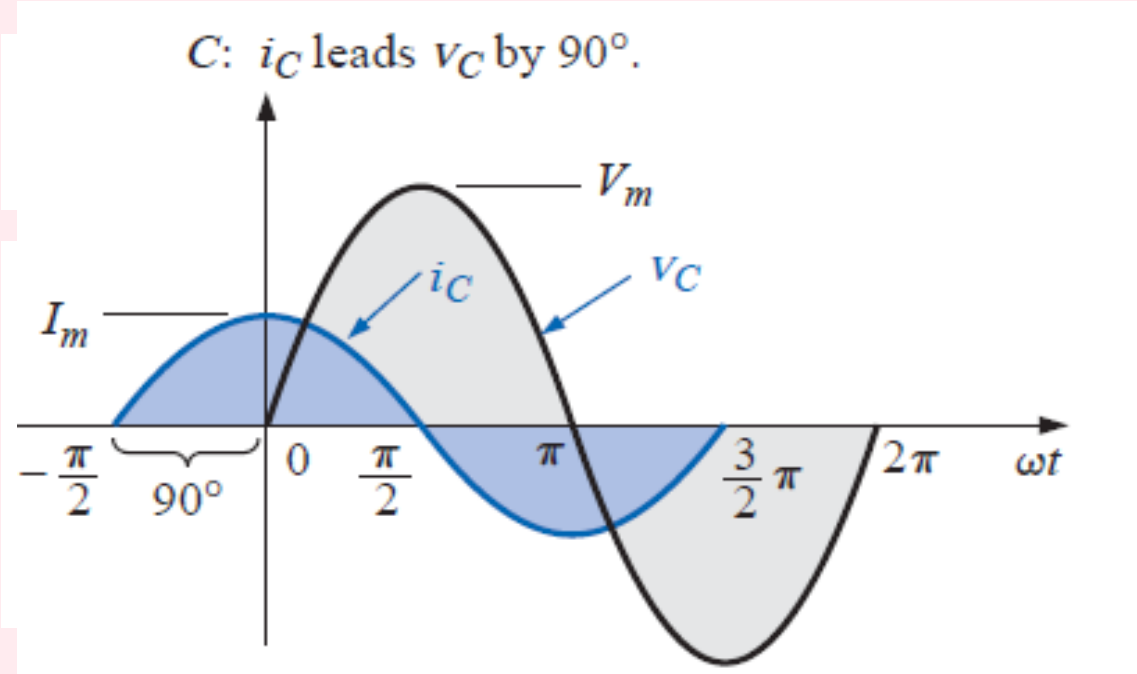

The quantity $\omega L$, called the **reactance** (from the wor
inductor, is symbolically represented by $X_L$ and is me

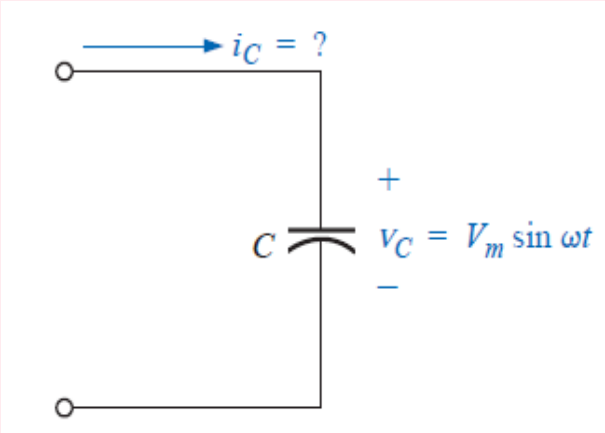
$$\boxed{X_L = \omega L}$$

Inductive reactance is the opposition to the flow of current, which results in the continual interchange of energy between the source and the magnetic field of the inductor. In other words, inductive reactance, unlike resistance (which dissipates energy in the form of heat), does not dissipate electrical energy (ignoring the effects of the internal resistance of the inductor).

# CAPACITOR

The current of a capacitor is directly related to the frequency and the capacitance of the capacitor. An increase in either quantity will result in an increase in the current of the capacitor.




for a capacitor, $\mathbf{i}_C$ leads $\mathbf{V}_C$ by 90°, or $\mathbf{V}_C$ lags $\mathbf{i}_C$ by 90°

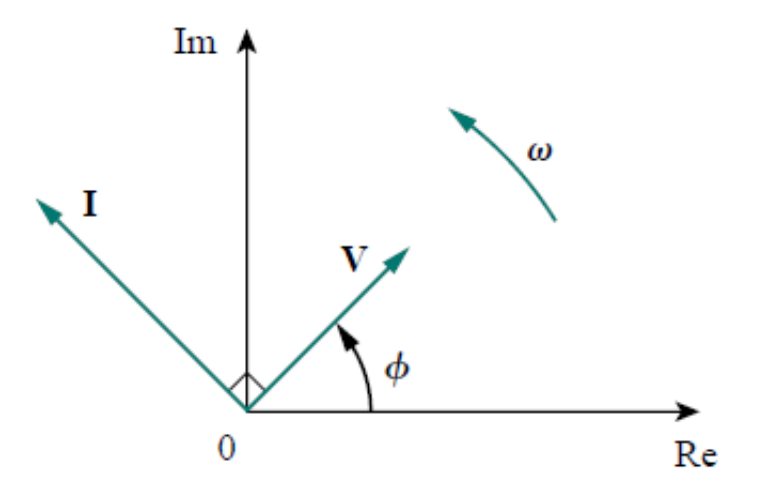
# CAPACITOR

If a phase angle is included in the sinusoidal expression for Vc, such as

$$v_C = V_m \sin(\omega t \pm \theta)$$

then

$$i_C = \omega C V_m \sin(\omega t \pm \theta + 90°)$$

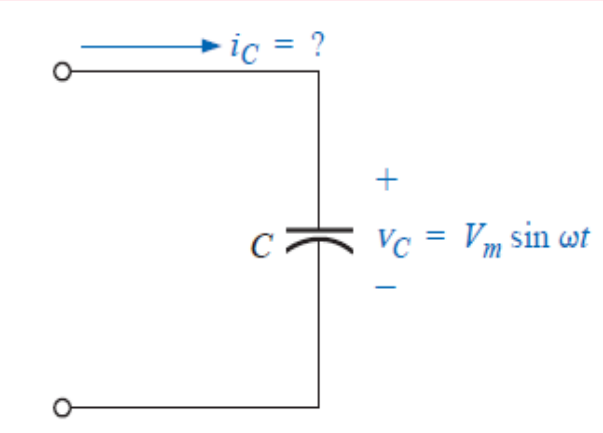



Phasor diagram for the capacitor I leads V.

# CAPACITOR

The opposition established by an capacitor in a sinusoidal ac network can now be found by

$$\text{Opposition} = \frac{\text{cause}}{\text{effect}} = \frac{V_m}{I_m} = \frac{V_m}{\omega C V_m} = \frac{1}{\omega C}$$




The quantity $1/\omega C$, called the **reactance** of a ca
cally represented by $X_C$ and is measured in ohms; t

$$X_C = \frac{1}{\omega C} \qquad \text{(ohms, } \Omega\text{)}$$

Capacitive reactance is the opposition to the flow of charge, which results in the continual interchange of energy between the source and the electric field of the capacitor. Like the inductor, the capacitor does not dissipate energy in any form (ignoring the effects of the leakage resistance).

Example: The current through a 0.1-H coil is provided. Find the sinusoidal expression for the voltage across the coil. Sketch the v and i curves.

a. $i = 10 \sin 377t$

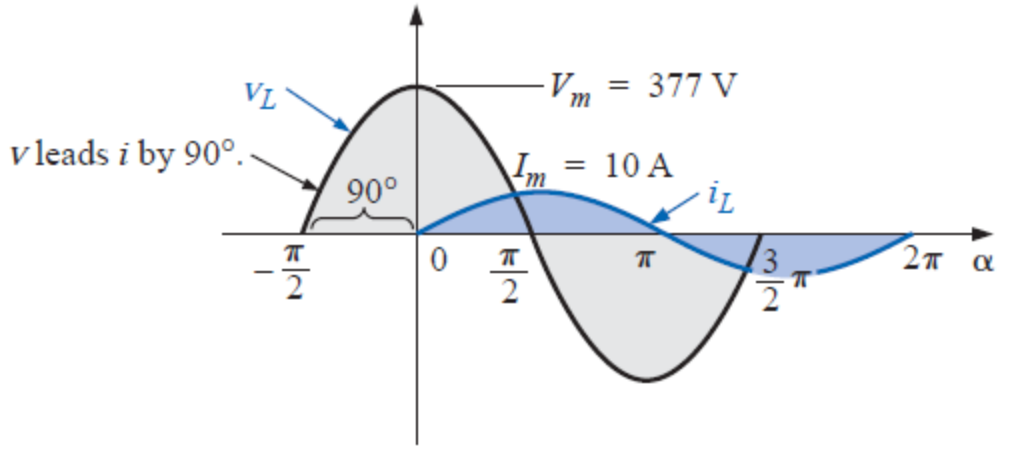
b. $i = 7 \sin(377t - 70°)$

**Solutions:**

$$X_L = \omega L = (377 \text{ rad/s})(0.1 \text{ H}) = 37.7\ \Omega$$

$$V_m = I_m X_L = (10 \text{ A})(37.7\ \Omega) = 377 \text{ V}$$

and we know that for a coil $v$ leads $i$ by 90°. Therefore,



b. $X_L$ remains at 37.7 Ω.

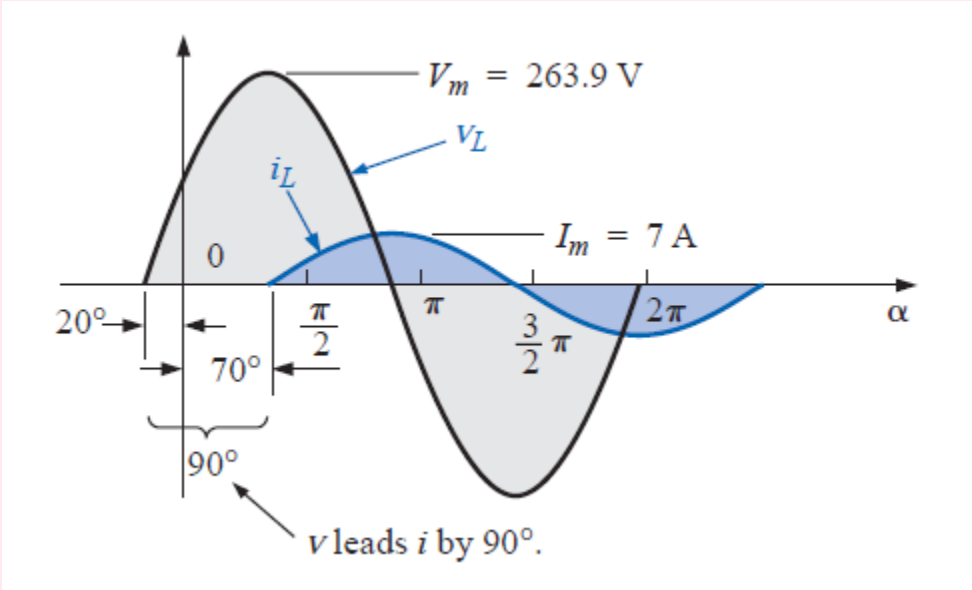
$$V_m = I_m X_L = (7\text{ A})(37.7\ \Omega) = 263.9\text{ V}$$

and we know that for a coil $v$ leads $i$ by 90°. Therefore,

$$v = 263.9 \sin(377t - 70° + 90°)$$

and

$$v = \mathbf{263.9 \sin(377t + 20°)}$$



Example: The current through a 100-μF capacitor is given. Find the sinusoidal expression for the voltage across the capacitor.

$$i = 40 \sin(500t + 60°)$$

**Solution:**

$$X_C = \frac{1}{\omega C} = \frac{1}{(500 \text{ rad/s})(100 \times 10^{-6} \text{ F})} = \frac{10^6 \ \Omega}{5 \times 10^4} = \frac{10^2 \ \Omega}{5} = 20 \ \Omega$$

$$V_m = I_m X_C = (40 \text{ A})(20 \ \Omega) = 800 \text{ V}$$

and we know that for a capacitor, $v$ lags $i$ by 90°. Therefore,

$$v = 800 \sin(500t + 60° - 90°)$$

and

$$v = \mathbf{800 \sin(500t - 30°)}$$

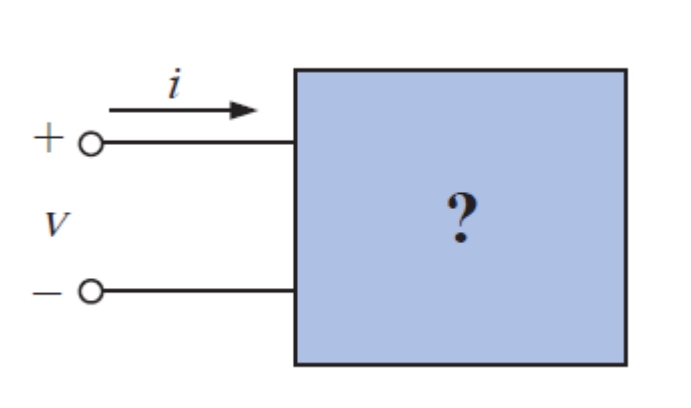
Example: For the following pairs of voltages and currents, determine whether the element involved is a capacitor, an inductor, or a resistor, and determine the value of C, L, or R if sufficient data are provided

a. $v = 100 \sin(\omega t + 40°)$
$i = 20 \sin(\omega t + 40°)$

b. $v = 1000 \sin(377t + 10°)$
$i = 5 \sin(377t - 80°)$

c. $v = 500 \sin(157t + 30°)$
$i = 1 \sin(157t + 120°)$

d. $v = 50 \cos(\omega t + 20°)$
$i = 5 \sin(\omega t + 110°)$



**Solutions:**

a. Since $v$ and $i$ are *in phase,* the element is a *resistor;* and

$$R = \frac{V_m}{I_m} = \frac{100\text{ V}}{20\text{ A}} = \mathbf{5\ \Omega}$$

b. Since $v$ *leads* $i$ by 90°, the element is an *inductor;* and

$$v = 1000 \sin(377t + 10°)$$
$$i = 5 \sin(377t - 80°)$$

$$X_L = \frac{V_m}{I_m} = \frac{1000\text{ V}}{5\text{ A}} = 200\ \Omega$$

so that $X_L = \omega L = 200\ \Omega$ or

$$L = \frac{200\ \Omega}{\omega} = \frac{200\ \Omega}{377\text{ rad/s}} = \mathbf{0.531\ H}$$

c. Since $i$ *leads* $v$ by 90°, the element is a *capacitor;* and

$$X_C = \frac{V_m}{I_m} = \frac{500\text{ V}}{1\text{ A}} = 500\ \Omega$$

$$v = 500 \sin(157t + 30°)$$
$$i = 1 \sin(157t + 120°)$$

so that $X_C = \dfrac{1}{\omega C} = 500\ \Omega$ or

$$C = \frac{1}{\omega 500\ \Omega} = \frac{1}{(157\text{ rad/s})(500\ \Omega)} = \mathbf{12.74\ \mu F}$$

d. $v = 50\cos(\omega t + 20°) = 50\sin(\omega t + 20° + 90°)$
$= 50\sin(\omega t + 110°)$

Since $v$ and $i$ are *in phase,* the element is a *resistor;* and

$$R = \frac{V_m}{I_m} = \frac{50\text{ V}}{5\text{ A}} = \mathbf{10\ \Omega}$$

$v = 50\cos(\omega t + 20°)$
$i = 5\sin(\omega t + 110°)$

Assignment

1. The voltage $v = 12\cos(60t + 45°)$ is applied to a 0.1-H inductor. Find the steady-state current through the inductor.

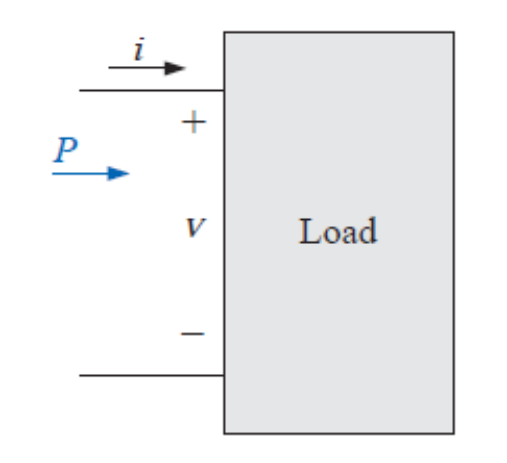
2. If voltage $v = 6\cos(100t - 30°)$ is applied to a 50 $\mu$F capacitor, calculate the current through the capacitor.

# AVERAGE POWER

For any load in a sinusoidal ac network, the voltage across the load and the current through the load will vary in a sinusoidal nature. If we take the general case use the following for v and i

$$v = V_m \sin(\omega t + \theta_v)$$

$$i = I_m \sin(\omega t + \theta_i)$$




then the power is defined by

$$p = vi = V_m \sin(\omega t + \theta_v) I_m \sin(\omega t + \theta_i)$$
$$= V_m I_m \sin(\omega t + \theta_v) \sin(\omega t + \theta_i)$$

Using the trigonometric identity

$$\sin A \sin B = \frac{\cos(A - B) - \cos(A + B)}{2}$$

# AVERAGE POWER

the function $\sin(\omega t + \theta_v)\sin(\omega t + \theta_i)$ becomes

$$\sin(\omega t + \theta_v)\sin(\omega t + \theta_i)$$

$$= \frac{\cos[(\omega t + \theta_v) - (\omega t + \theta_i)] - \cos[(\omega t + \theta_v) + (\omega t + \theta_i)]}{2}$$

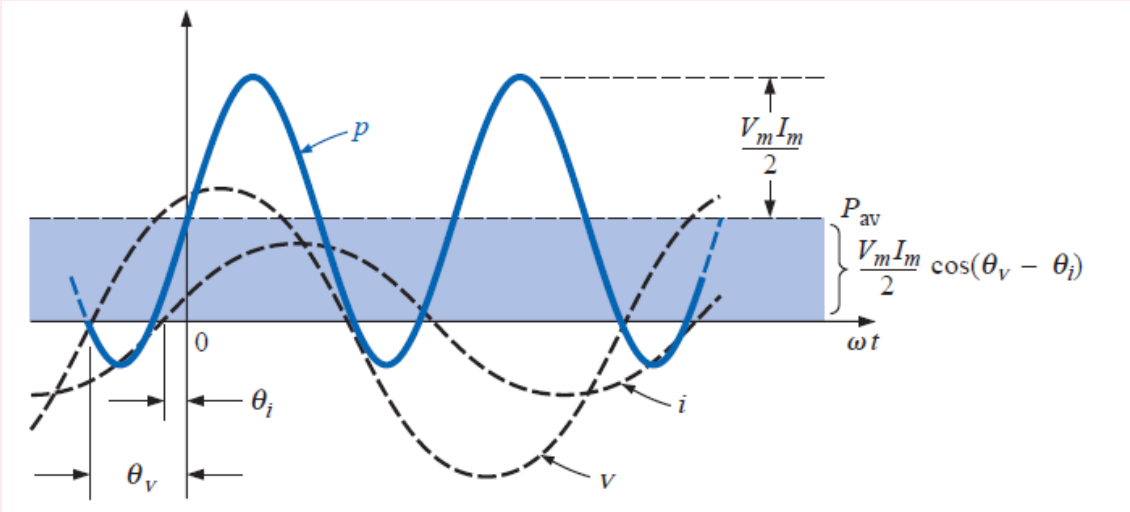
$$= \frac{\cos(\theta_v - \theta_i) - \cos(2\omega t + \theta_v + \theta_i)}{2}$$

so that

$$p = \overbrace{\left[\frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\right]}^{\text{Fixed value}} - \overbrace{\left[\frac{V_m I_m}{2}\cos(2\omega t + \theta_v + \theta_i)\right]}^{\text{Time-varying (function of } t)}$$

# AVERAGE POWER

A plot of v, i, and p on the same set of axes is shown in following Fig.




The first term in the preceding equation, however, has a constant magnitude (no time dependence) and therefore provides some net transfer of energy. The **average power**, or **real power** as it is sometimes called, is the power delivered to and dissipated by the load.

the magnitude of average power delivered is independent of whether v leads i or i leads v.

# AVERAGE POWER

Defining $\theta$ as equal to $|\theta_v - \theta_i|$, where $|\quad|$ indicates that only the magnitude is important and the sign is immaterial, we have

$$\boxed{P = \frac{V_m I_m}{2} \cos \theta} \qquad \text{(watts, W)}$$

where $P$ is the average power in watts. This equation can also be written

$$P = \left(\frac{V_m}{\sqrt{2}}\right)\left(\frac{I_m}{\sqrt{2}}\right) \cos \theta$$

or, since $\quad V_{\text{eff}} = \dfrac{V_m}{\sqrt{2}} \quad \text{and} \quad I_{\text{eff}} = \dfrac{I_m}{\sqrt{2}}$

$$\boxed{P = V_{\text{eff}} I_{\text{eff}} \cos \theta}$$

# AVERAGE POWER

## Resistor

In a purely resistive circuit, since $v$ and $i$ are in phase, $|\theta_v - \theta_i| = \theta = 0°$, and $\cos\theta = \cos 0° = 1$, so that

$$\boxed{P = \frac{V_m I_m}{2} = V_{\text{eff}} I_{\text{eff}}} \qquad \text{(W)}$$

Or, since

$$I_{\text{eff}} = \frac{V_{\text{eff}}}{R}$$

then

$$\boxed{P = \frac{V_{\text{eff}}^2}{R} = I_{\text{eff}}^2 R} \qquad \text{(W)}$$

# AVERAGE POWER

## Inductor

In a purely inductive circuit, since $v$ leads $i$ by 90°, $|\theta_v - \theta_i| = \theta = |-90°| = 90°$. Therefore,

$$P = \frac{V_m I_m}{2} \cos 90° = \frac{V_m I_m}{2}(0) = \mathbf{0\ W}$$

***The average power or power dissipated by the ideal inductor (no associated resistance) is zero watts.***

# AVERAGE POWER

## Capacitor

In a purely capacitive circuit, since $i$ leads $v$ by 90°, $|\theta_v - \theta_i| = \theta = |-90°| = 90°$. Therefore,

$$P = \frac{V_m I_m}{2}\cos(90°) = \frac{V_m I_m}{2}(0) = \mathbf{0\ W}$$

***The average power or power dissipated by the ideal capacitor (no associated resistance) is zero watts.***

Example: Determine the average power delivered to networks having the following input voltage and current

$$v = 100 \sin(\omega t + 40°)$$
$$i = 20 \sin(\omega t + 70°)$$

$$V_m = 100, \qquad \theta_v = 40°$$
$$I_m = 20, \qquad \theta_i = 70°$$
$$\theta = |\theta_v - \theta_i| = |40° - 70°| = |-30°| = 30°$$

and

$$P = \frac{V_m I_m}{2} \cos\theta = \frac{(100\text{ V})(20\text{ A})}{2} \cos(30°) = (1000\text{ W})(0.866)$$
$$= \mathbf{866\ W}$$

Assignment : Determine the average power delivered to networks having the following input voltage and current

$$v = 150 \sin(\omega t - 70°)$$
$$i = 3 \sin(\omega t - 50°)$$

# POWER FACTOR

It can be defined as the absolute value of the cosine of the phase shift between the voltage and current in an AC circuit.

$$\text{Power factor} = F_p = \cos\theta$$

The terms leading and lagging are often written in conjunction with the power factor. They are defined by the current through the load. If the current leads the voltage across a load, the load has a leading power factor. If the current lags the voltage across the load, the load has a lagging power factor.

capacitive networks have leading power factors, and inductive networks have lagging power factors.